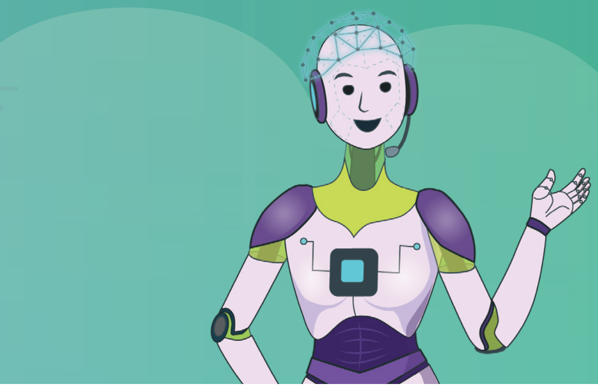
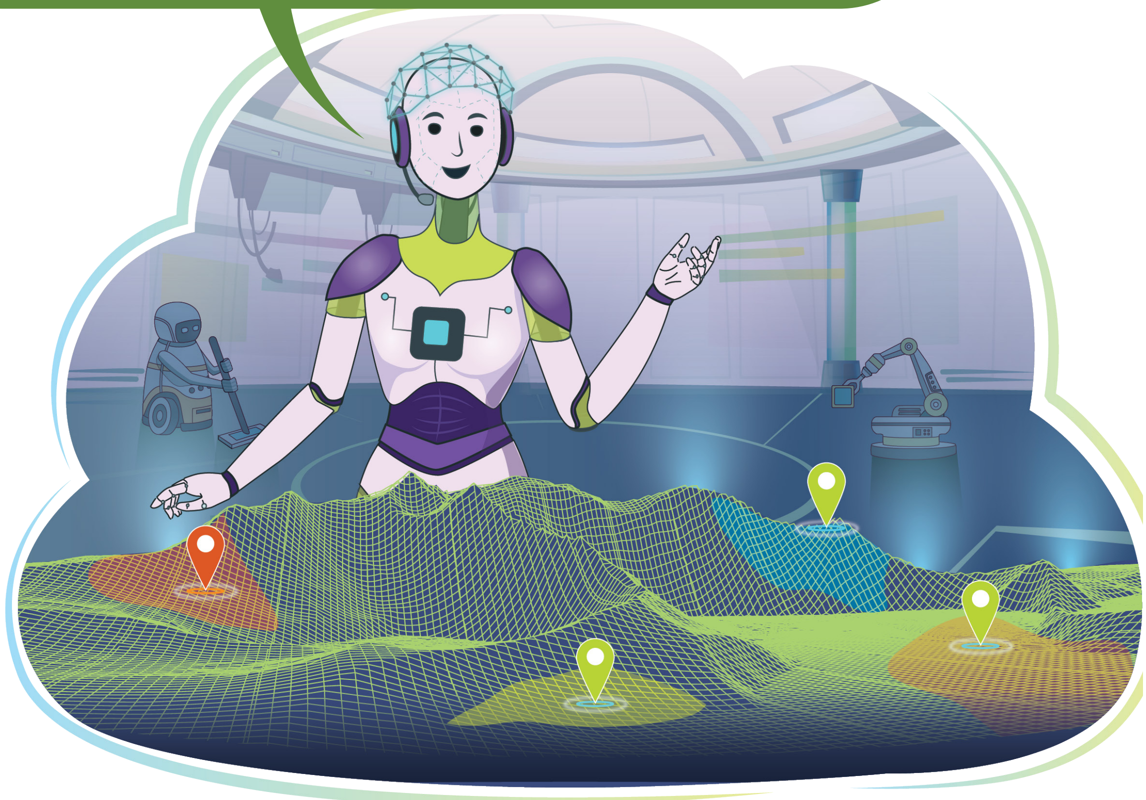


Introduction to Augmented Reality

Did you know that Augmented Reality (AR) technology was initially developed for military purposes? AR enhanced soldiers' situational awareness by overlaying crucial information on their field of view. Today, AR has found its way into various fields, such as gaming, education, and healthcare, making it a versatile and innovative technology!



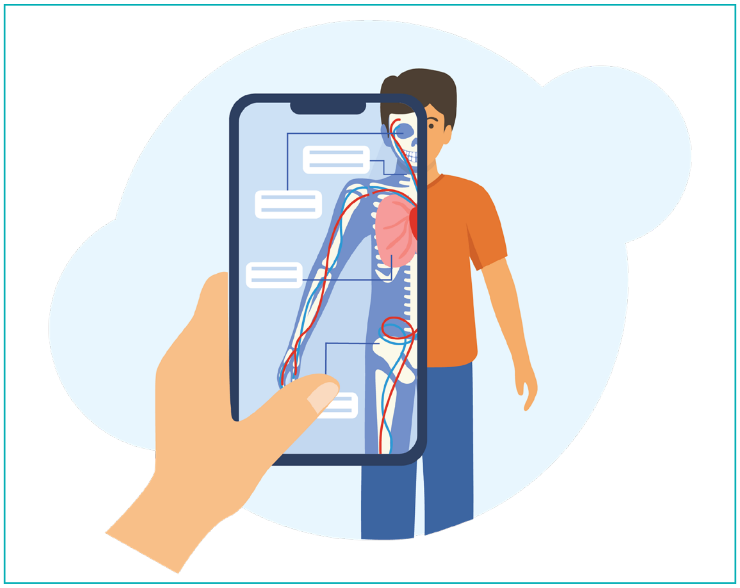
In this era of the 'Digital Age' or 'Information Age', the amount of digital information/data available to us is huge. However, there is a significant gap between this digital world of data and the physical world that we live in. Our world is three-dimensional, while the screens on which we access the plethora of digital information are two-dimensional. This gap limits us from utilising the power of the digital world. Therefore, to bridge this gap, an exciting set of technologies is emerging, known as **Extended Reality.**

Extended Reality

Extended Reality or **XR,** is an umbrella term for all the available immersive technologies such as AR (Augmented Reality), VR (Virtual Reality), and MR (Mixed Reality).

Let us understand the differences between these technologies –

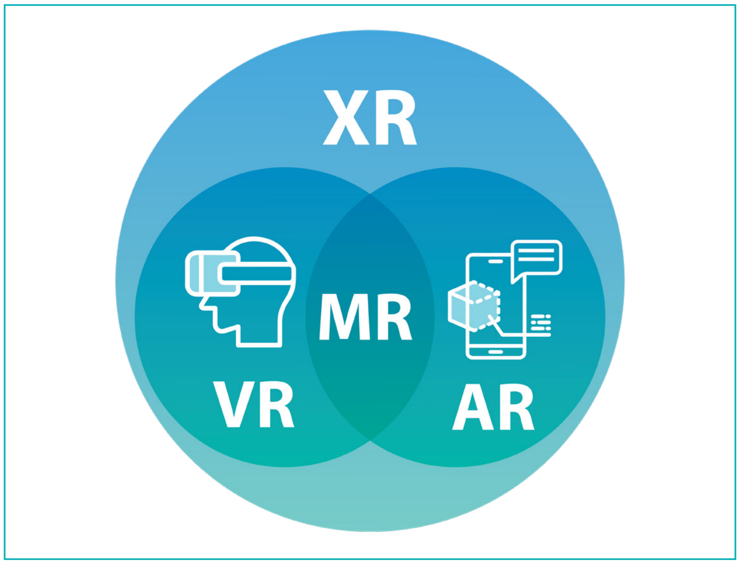
Augmented Reality (AR): AR overlays virtual objects in our real-world environment. It enriches our perception and enhances our experiences.



Virtual Reality (VR): It provides the user with a feeling of being present in a completely new environment. VR makes use of computer technology to create a simulated environment that can be explored in 360 degrees.

Mixed Reality (MR): Mixed Reality, as the name suggests, is a mixture of both AR and VR. In MR, a user can interact with both the real world and the virtual environment.

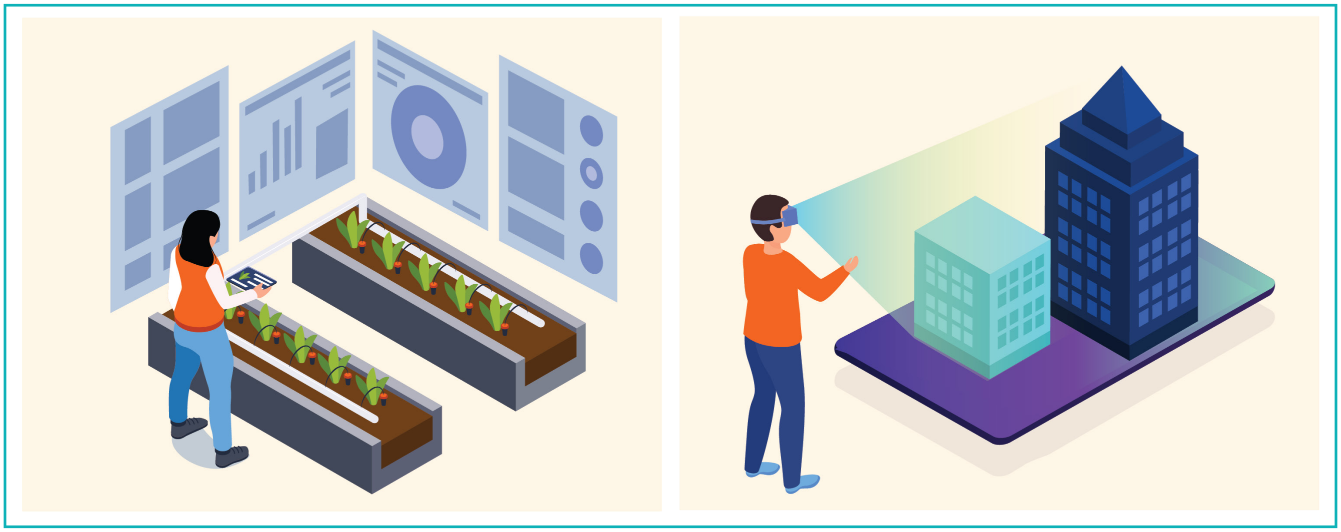
The relationship between different XR terms can be understood through a Venn diagram shown below.



Today, we are going to create an Augmented Reality (AR) application, like the one that we have seen on Google.com, where we can view a 3D animal/bird in our surroundings.

Augmented Reality (AR) refers to a technology that combines computer-generated elements with the real-world environment to create an immersive experience for the user. AR overlays digital information such as images, videos, 3D models, or text, onto the real world, typically using devices such as smartphones, tablets, smart glasses, or headsets.

The applications of AR are extensive. It can be used to display additional information about physical objects, provide step-by-step instructions for tasks, offer interactive educational experiences, enable virtual try-on for shopping, facilitate remote collaboration, enhance training and simulations, and even transform the way we play games and engage with entertainment content.



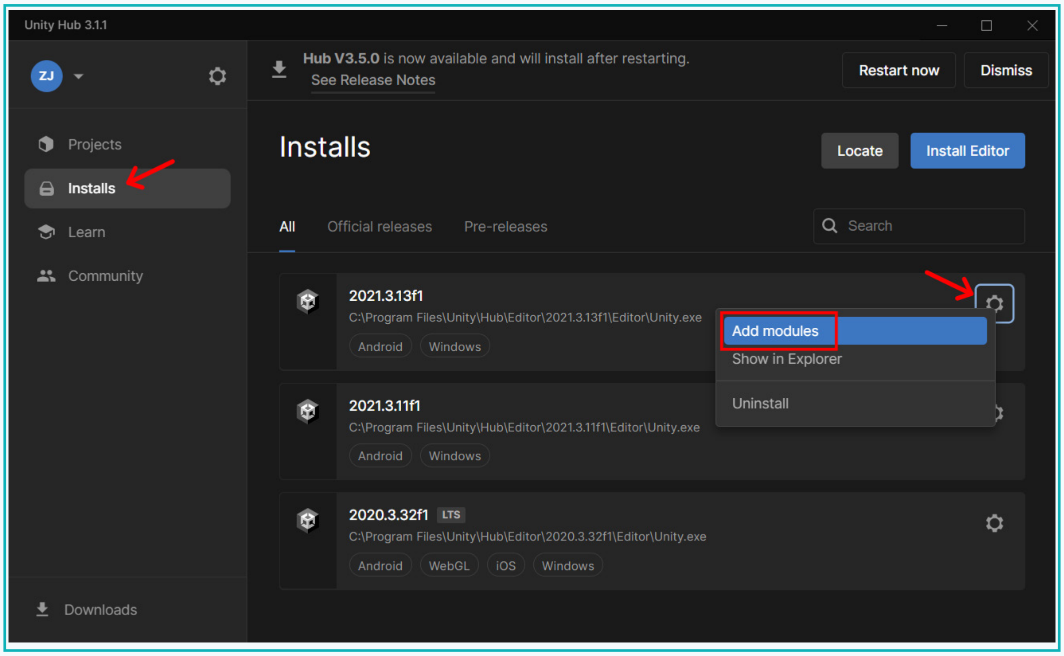
Installation of Required Packages

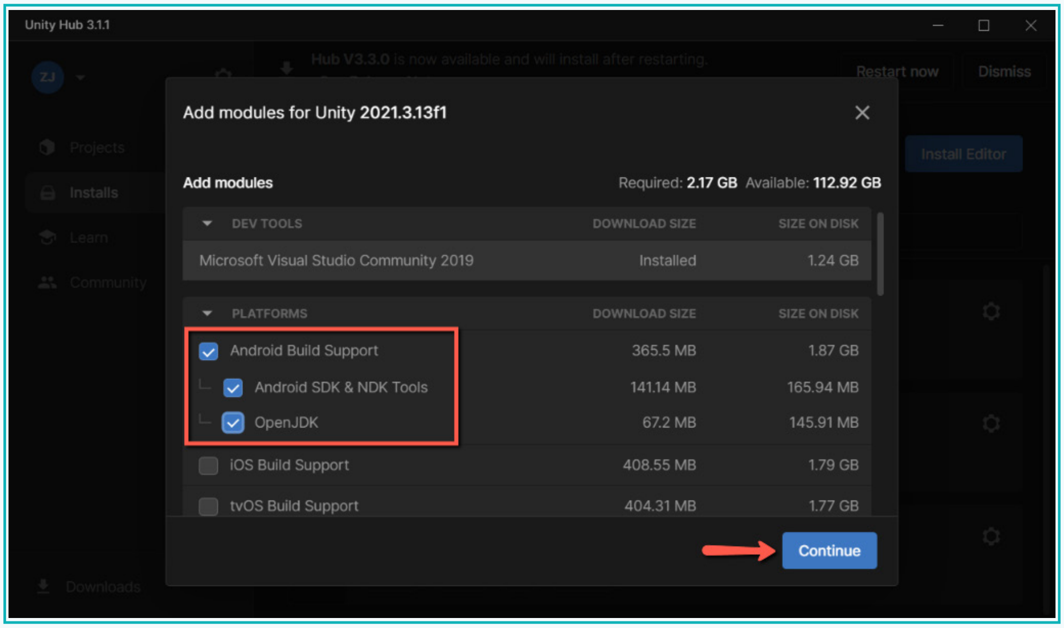
To publish XR-based applications to the Android platform, we need to install the Android build support in Unity. The Android Build Support provides functionality to create applications specifically designed for Android devices.

During installation, Unity provides an option for installing these modules. However, if we have not done so during installation, we can also add these modules to an already installed Unity Editor version.

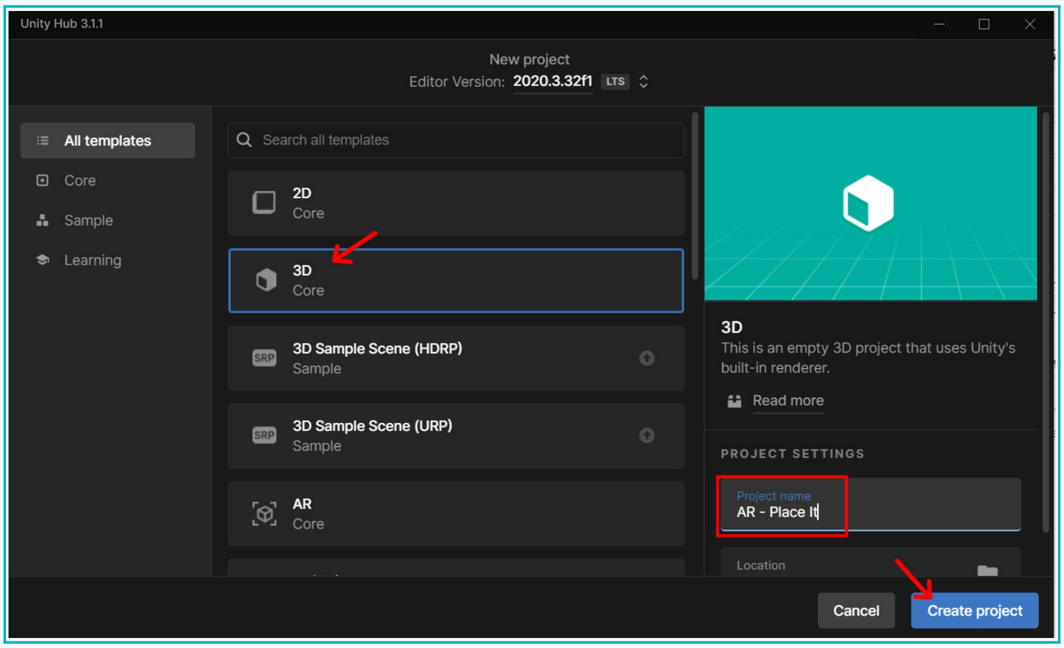
Activity

1. Use the Unity Hub to install Android Build Support and the associated dependencies, which include the Android SDK & NDK tools, as well as OpenJDK. In the 'Installs' tab, click on the 'gear' icon next to the Unity version we plan to use for our AR Application. Check the 'Android Build Support' module and click on 'Continue'.



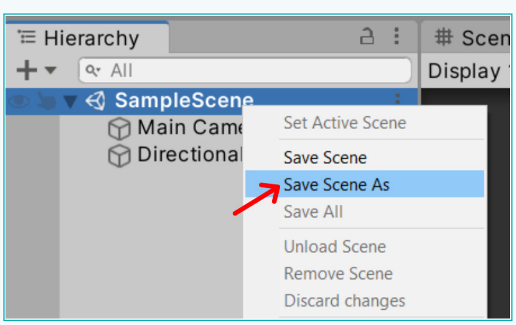


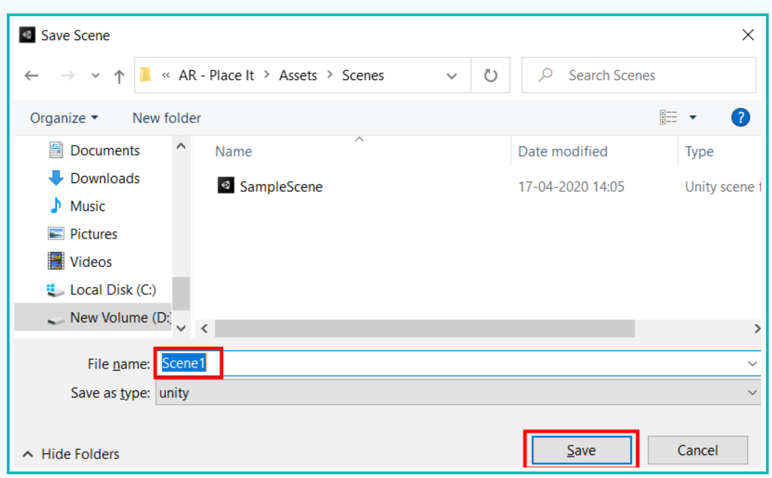
2 Once the selected modules are installed, click on the Projects tab in Unity Hub. Select ‘3D’ from the ‘Template’ list. Create a new project named “AR – Place It” and set its location. Click on ‘Create Project’.



The Unity Editor will open.

3 Rename the scene.





Next, we will install the **AR Foundation** package.

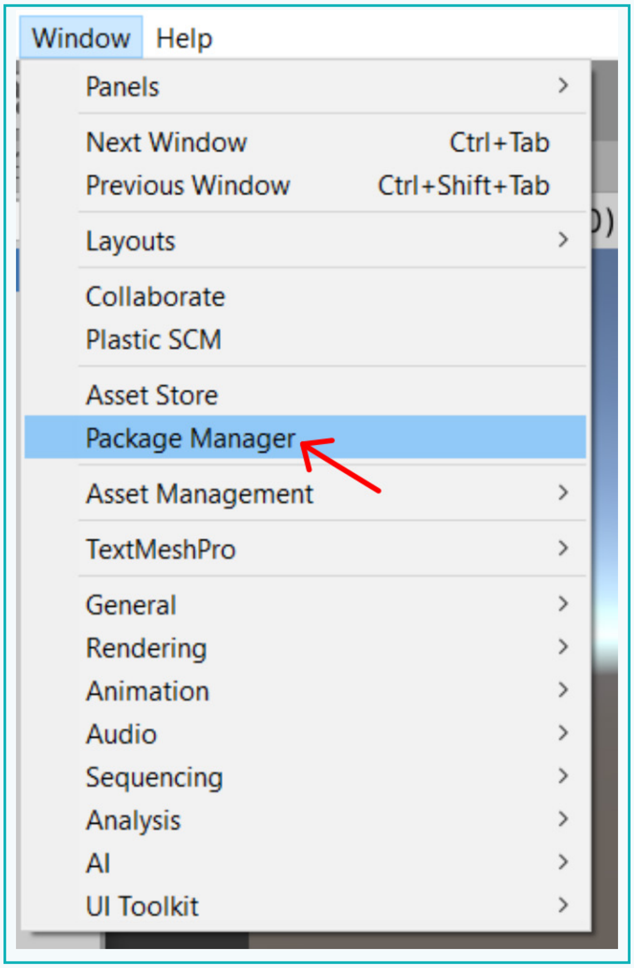
The **AR Foundation** package is required in Unity AR projects as it enables developers to create AR experiences that are compatible with multiple AR platforms. This means that using AR Foundation, we can build an AR Unity Application and deploy it to either an iOS or an Android device without having to write a separate code for each platform. It also exposes platform-specific features, when needed.

AR Foundation handles device tracking, which is crucial for creating accurate and stable AR experiences. It provides access to the device's camera, motion sensors, and other hardware components required to enable real-time tracking of the user's position and orientation. AR Foundation also supports features such as plane detection, object tracking, light estimation, and spatial mapping, all of which are essential for creating interactive and immersive AR applications.

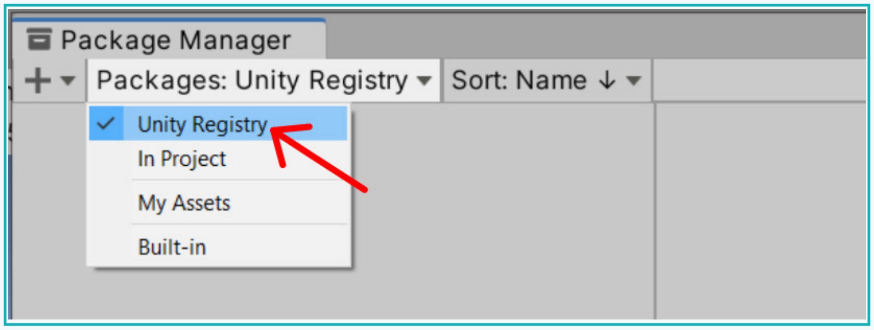
The AR Foundation package is critical to include when building an AR Unity application.

We will import this package into Unity using the Package Manager. The **Package Manager** helps us in importing Unity-verified packages into Unity for us to use in our applications.

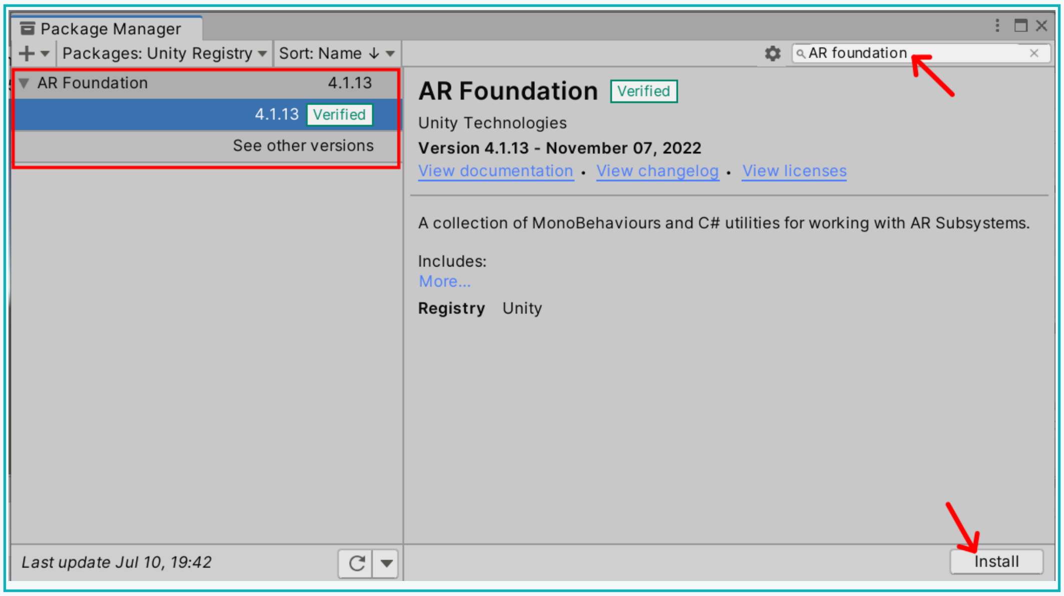
4. Click on 'Package Manager' under the 'Windows' tab.



5 Select 'Unity Registry', as shown.



6 In the Search bar, type 'AR Foundation', then select the 'AR Foundation package' from the list, and click 'Install'.



In addition to installing AR Foundation, we also require the **ARCore** or the **ARKit** package.

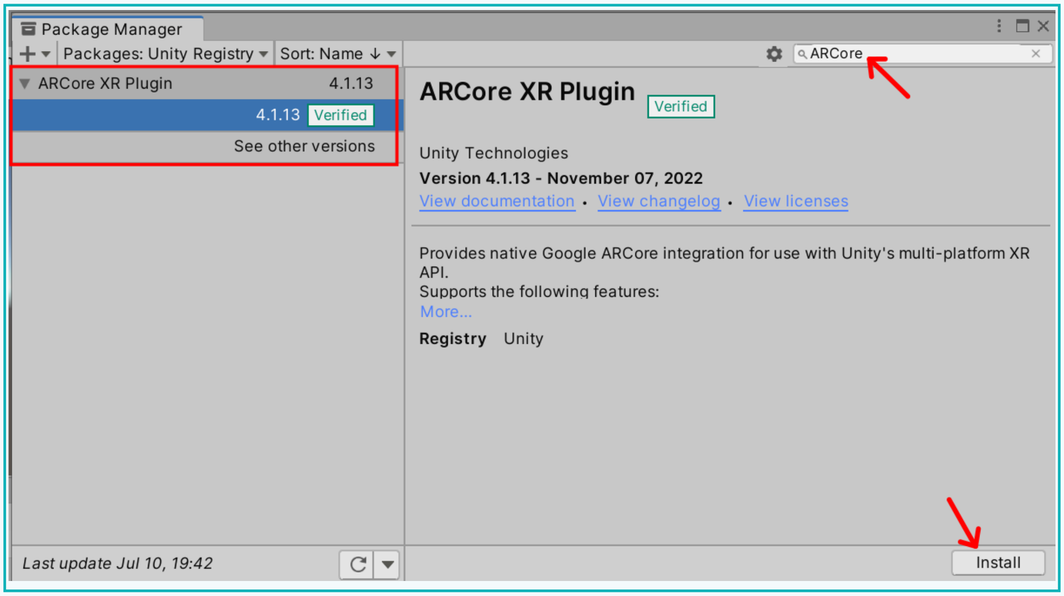
The **ARCore** and **ARKit** packages provide the underlying AR platform-specific functionalities and capabilities for Android and iOS devices, respectively.

While AR Foundation is the unifying framework for all AR Applications, to leverage the specific features and optimizations offered by each platform, we need to make use of the ARCore/ARKit components.

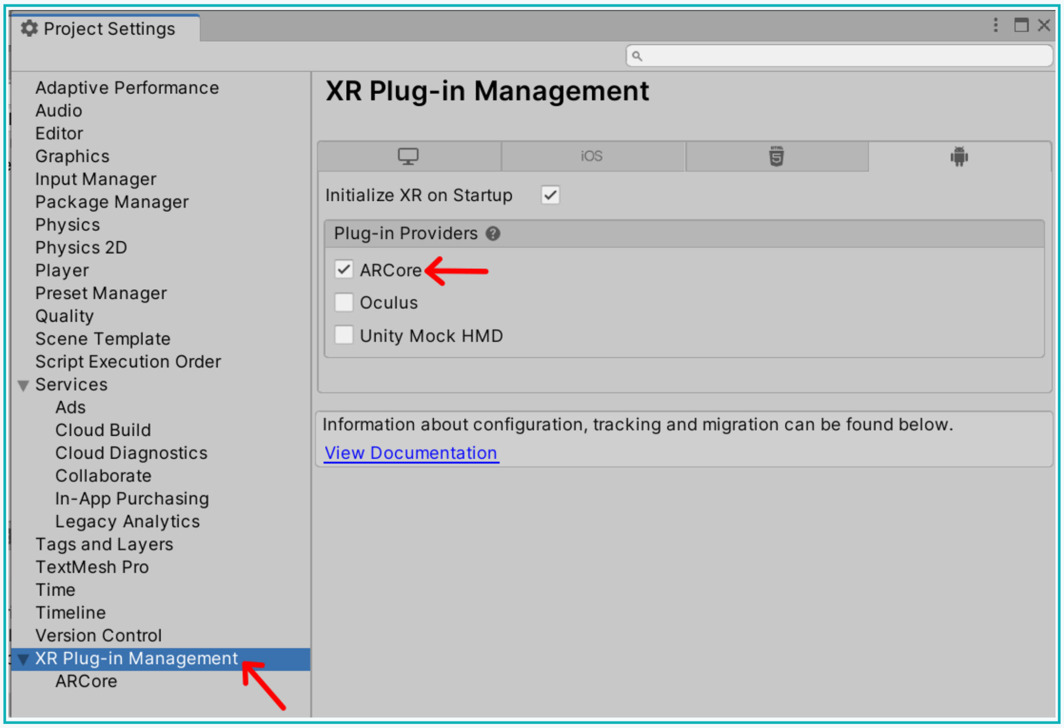
ARCore is developed by **Google** and **ARKit** is developed by **Apple**. They provide advanced tracking capabilities and features tailored to their respective platforms. ARCore and ARKit are maintained and updated respectively by Google and Apple to ensure compatibility with the latest device models and operating system versions.

Since we are deploying to Android, we need to install the 'ARCore XR Plugin' package from the Unity Package Manager.

7 Type 'ARCore XR Plugin' in the search bar and install the plugin.



8 Navigate to 'Edit' → 'Project Settings'. Click on 'XR Plug-in Management' and under the 'Android' tab, enable 'ARCore'.



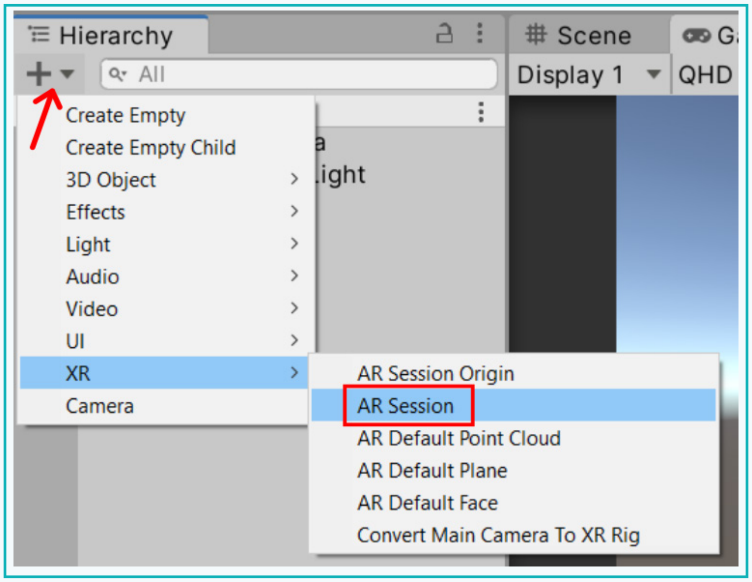
Now that all our AR packages are installed, we can put them to use.

Set Up the Scene for AR

AR Session:

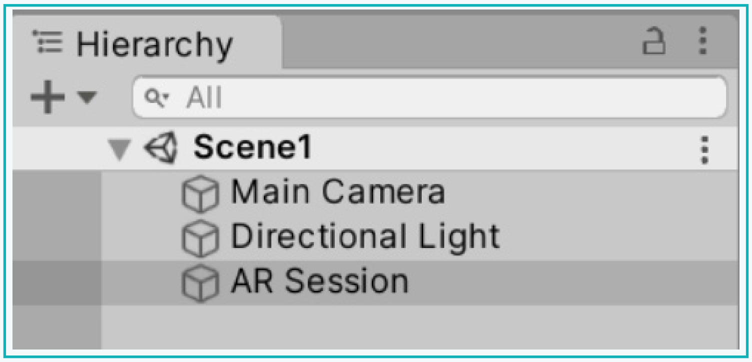
AR session helps us track AR processes in our scene such as motion tracking, environmental understanding, lighting estimation and overall management of the augmented reality experience. It controls the lifecycle of an AR experience.

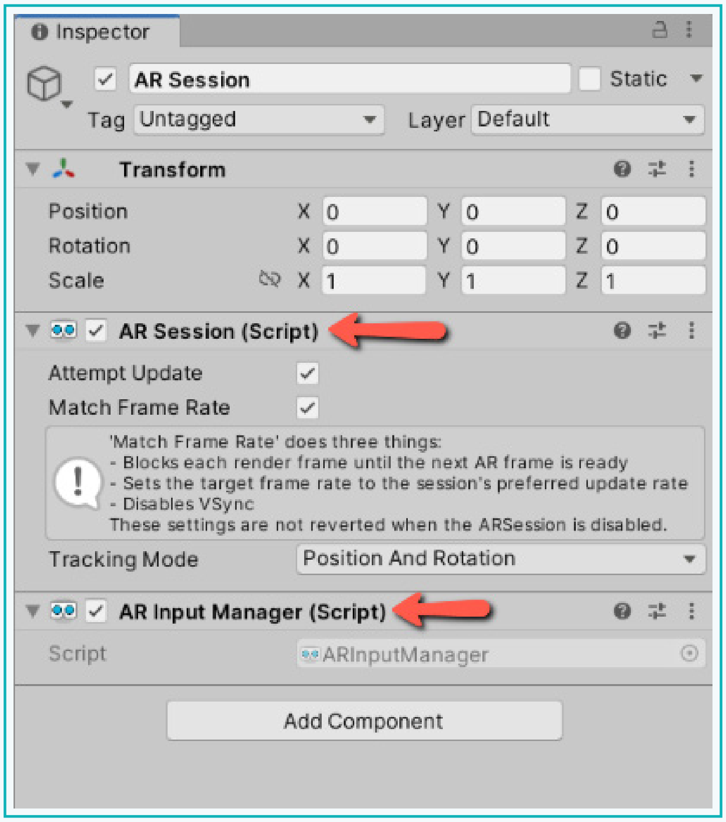
1. Add an ‘AR Session’ gameobject (‘XR’ → ‘AR Session’).



The AR Session is essential for AR to operate, but we do not need to interact with it much. It controls the augmented reality configuration options while our app is open. It also manages the lifecycle of an AR experience, enabling or disabling AR on the target platform. If the scene does not contain an ‘ARSession’ gameobject, the application will not be able to track features in its environment.

The ‘AR Session’ gameobject includes both an ‘AR Session’ component and an ‘AR Input Manager’ component.





AR Session Origin

The **AR Session Origin** is a crucial component that acts as a bridge between the real and virtual worlds when developing Augmented Reality (AR) applications. It serves as the anchor point for AR content within the Unity scene.

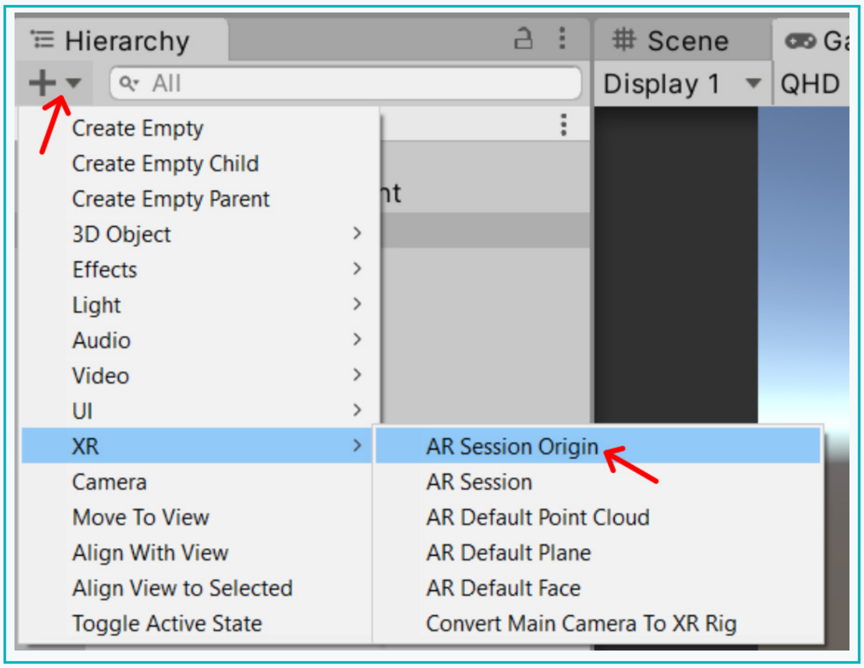
The 'AR Session Origin' is responsible for managing the AR session and tracking the device's position and orientation in the real world. It interacts with the AR tracking system provided by the AR platform (such as ARKit for iOS or ARCore for Android) to gather information about the user's environment and device motion.

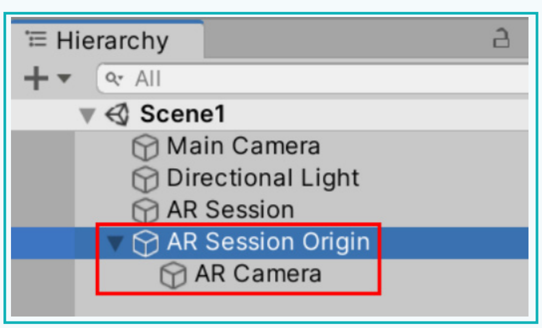
It provides methods for placing virtual objects in the correct location with respect to the real-world environment. It handles the translation, rotation, and scaling of virtual objects to match the physical world.

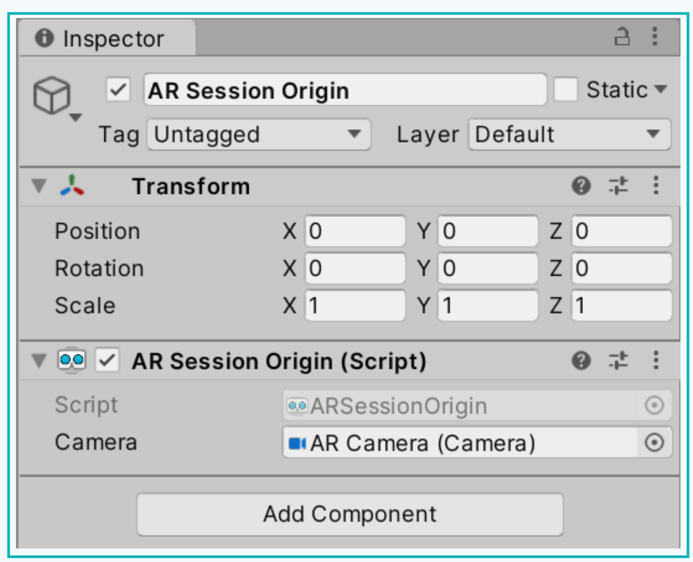
AR Session Origin also facilitates the intersection of virtual rays (straight lines) with real-world objects or surfaces. It helps in interaction between the virtual and real world, and object placement, thus providing a realistic and immersive experience. It serves as the foundation for most of our interactivity and functionality in AR.

2 Add an ‘AR Session Origin’ gameobject (‘XR’ → ‘AR Session Origin’).

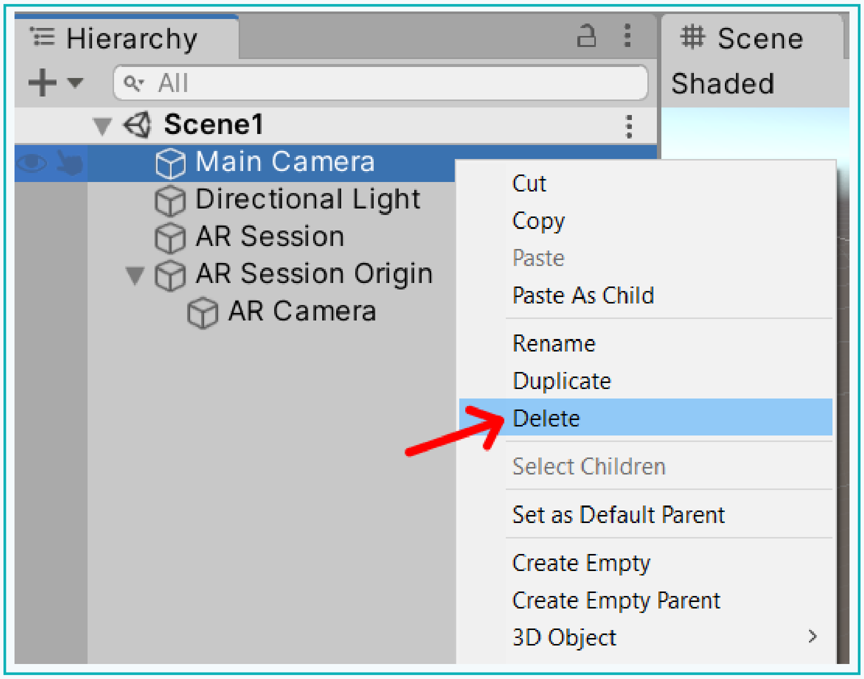
The AR Session Origin has an ‘AR Camera’ as a sub-component.





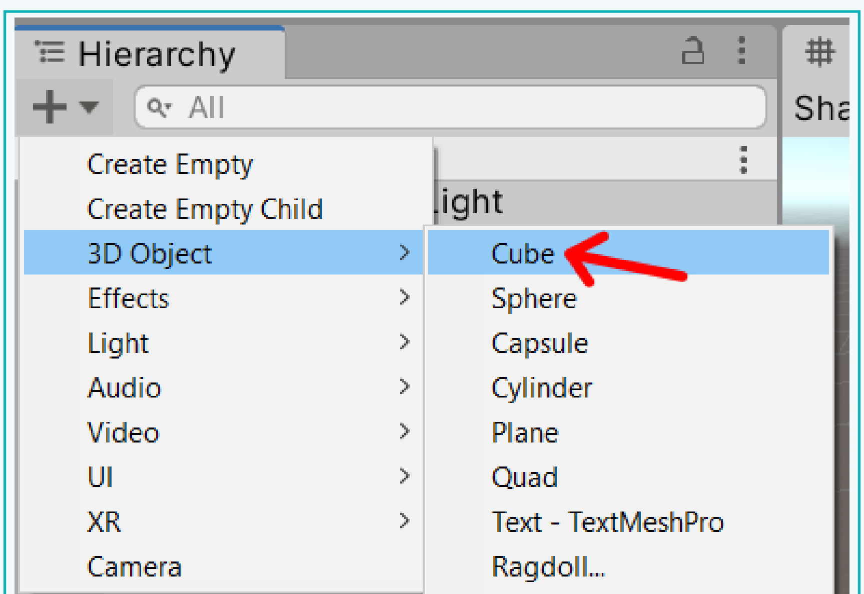


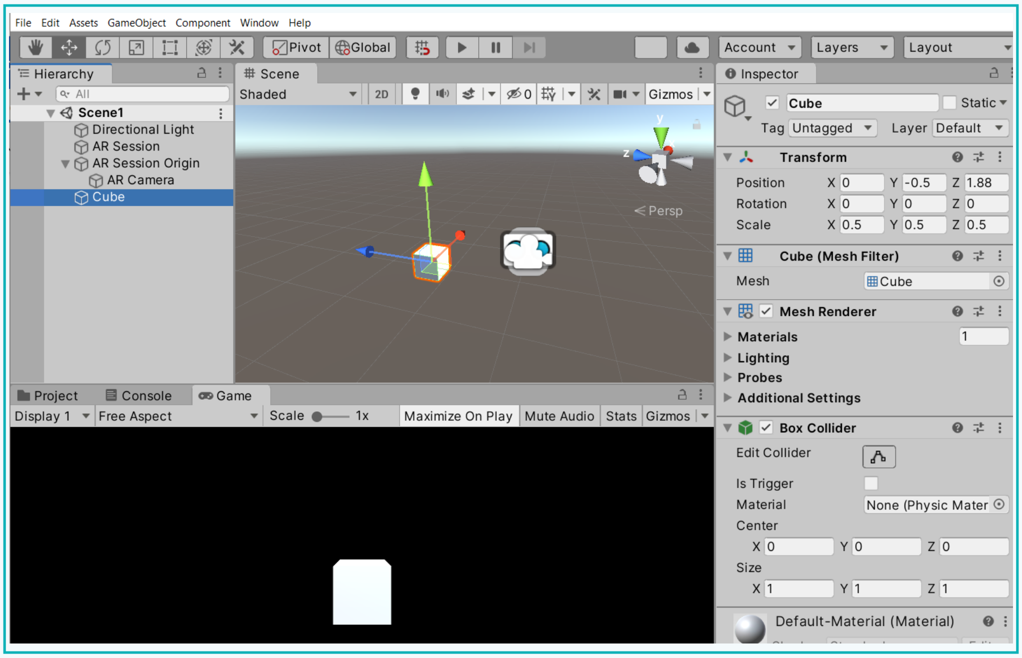
It is an AR camera that is configured for our AR application; hence, we can now delete the default 'Main Camera' that is already present in the scene.



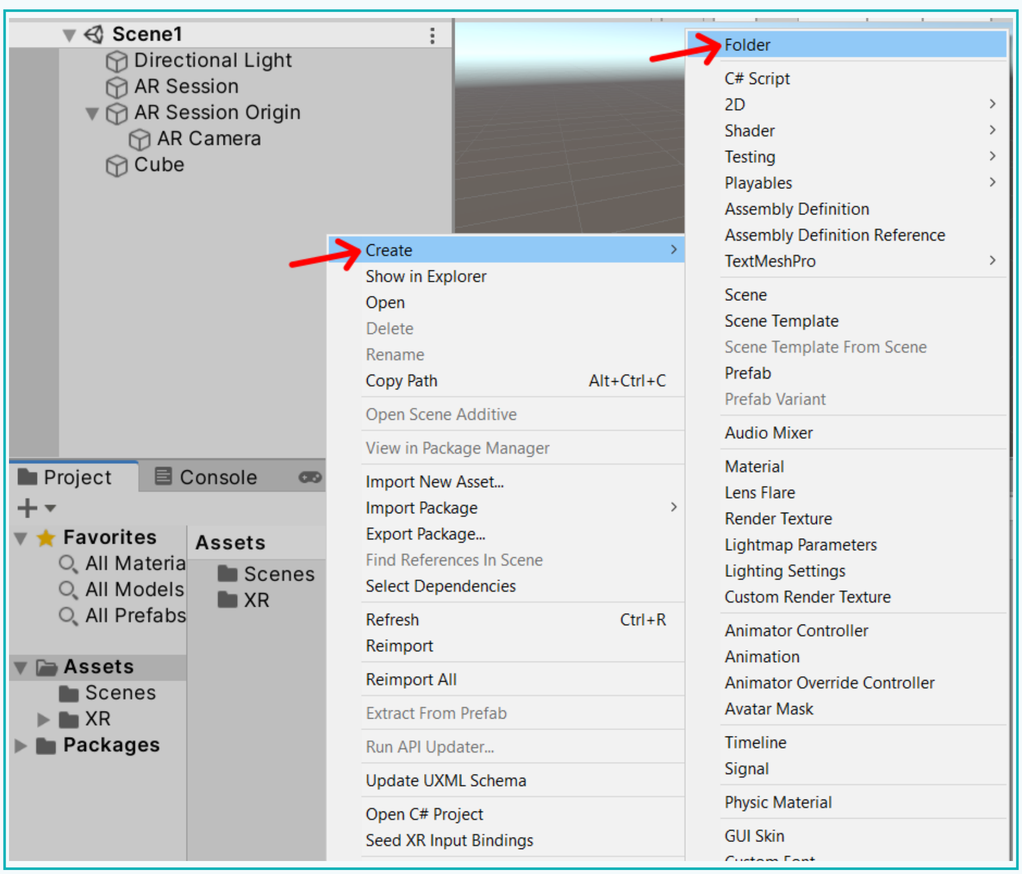
We have seen all the primary AR components that are required to make an AR experience. Now, we are ready to add objects to our scene.

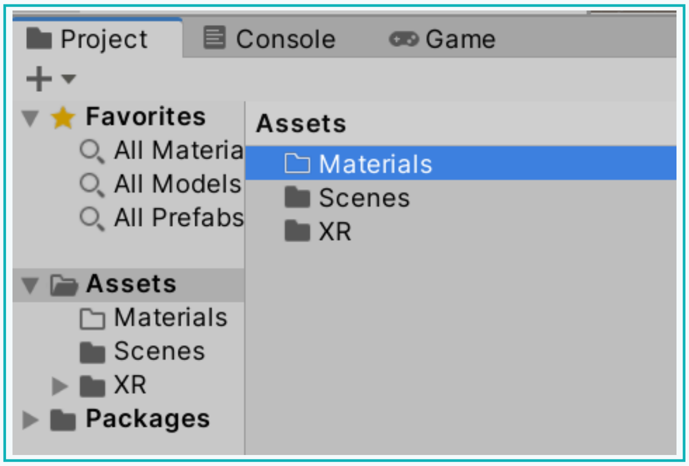
3. Let us create a new 'Cube' gameobject. Scale it down to (0.5, 0.5, 0.5). Position it such that it is visible in the 'Game' view.





4. Create a new folder ‘Materials’. Within it, create a new material and assign it any colour of your choice.




Project
Console
Game
Favorites
All Materia
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Assets
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Scenes
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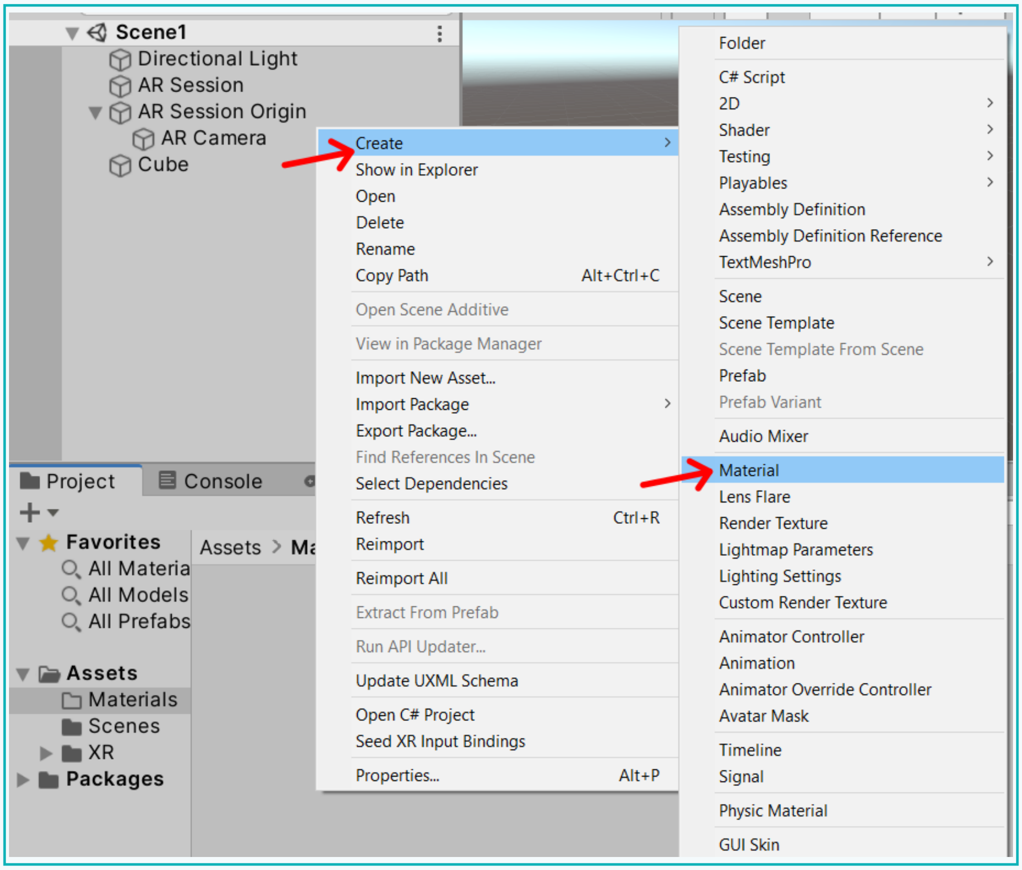
Scene1
Directional Light
AR Session
AR Session Origin
AR Camera
Cube
Create
Show in Explorer
Open
Delete
Rename
Copy Path Alt+Ctrl+C
Open Scene Additive
View in Package Manager
Import New Asset...
Import Package
Export Package...
Find References In Scene
Select Dependencies
Refresh Ctrl+R
Reimport
Reimport All
Extract From Prefab
Run API Updater...
Update UXML Schema
Open C# Project
Seed XR Input Bindings
Properties... Alt+P
Folder
C# Script
2D
Shader
Testing
Playables
Assembly Definition
Assembly Definition Reference
TextMeshPro
Scene
Scene Template
Scene Template From Scene
Prefab
Prefab Variant
Audio Mixer
Material
Lens Flare
Render Texture
Lightmap Parameters
Lighting Settings
Custom Render Texture
Animator Controller
Animation
Animator Override Controller
Avatar Mask
Timeline
Signal
Physic Material
GUI Skin
Project
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Packages
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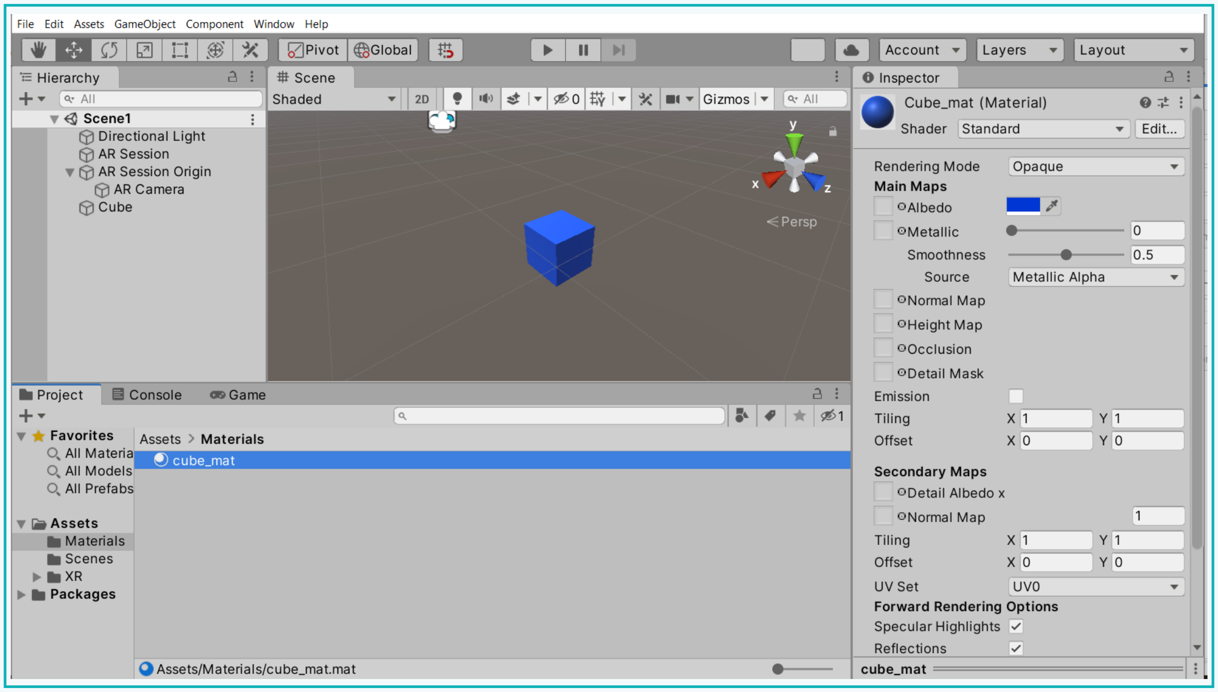
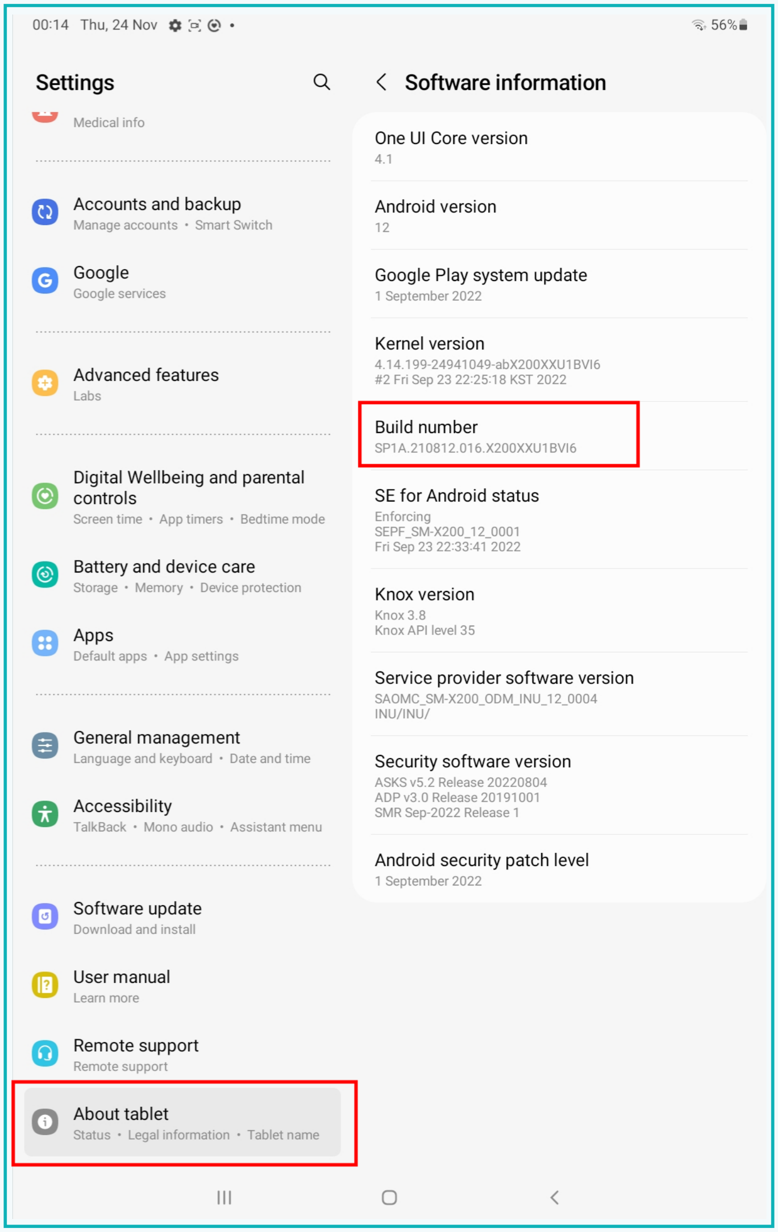


Publish to an Android Device

We have set up a basic scene for AR. Now, to test it on an Android device, we'll publish it. Before proceeding, certain settings must be configured to establish communication between our Android device and the computer/laptop containing our Unity AR project.

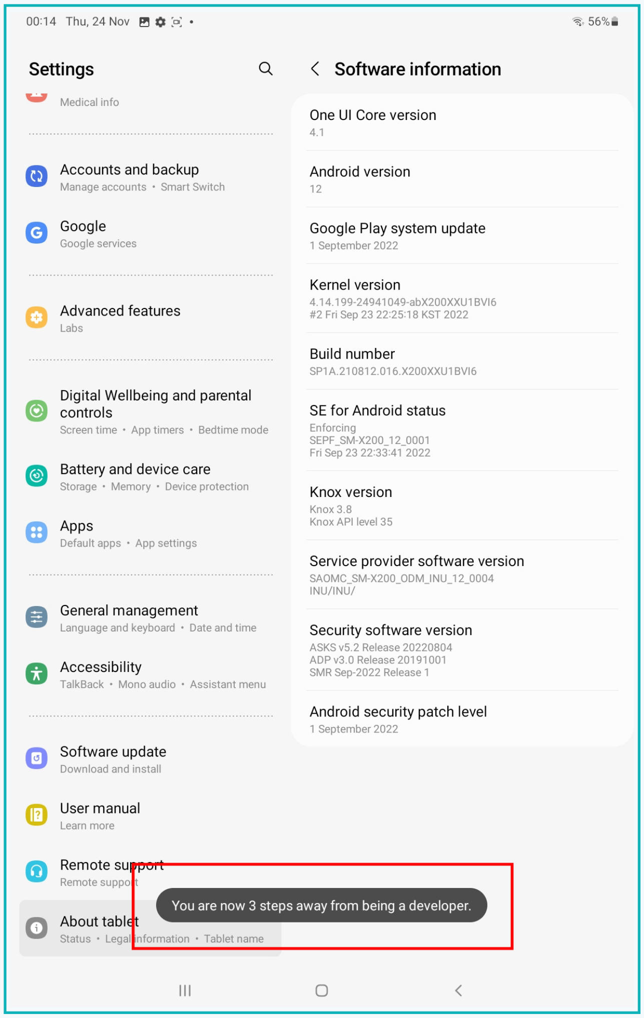
Firstly, ensure that the Android device is connected to the laptop/computer using a USB cable. Next, we need to enable the 'Developer' options and 'USB debugging' on the Android device.

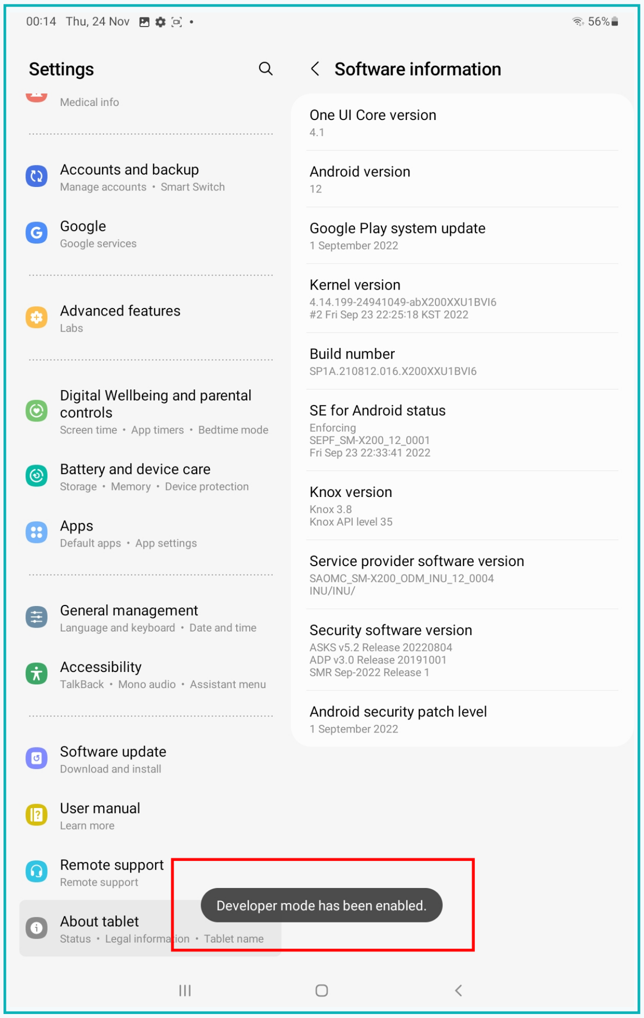
1 To enable 'USB debugging', you must enable the 'Developer' options on your device. To do this, find the build number in your device's 'Settings' menu. The location of the build number varies between devices, although it is usually present in the path: 'Settings' → 'About phone'/'About Tablet' → 'Build number'.



Since this was tested on a tablet, the screenshot shows the option - 'About Tablet'. The 'About Phone' option will be displayed if the project is tested on a mobile device.

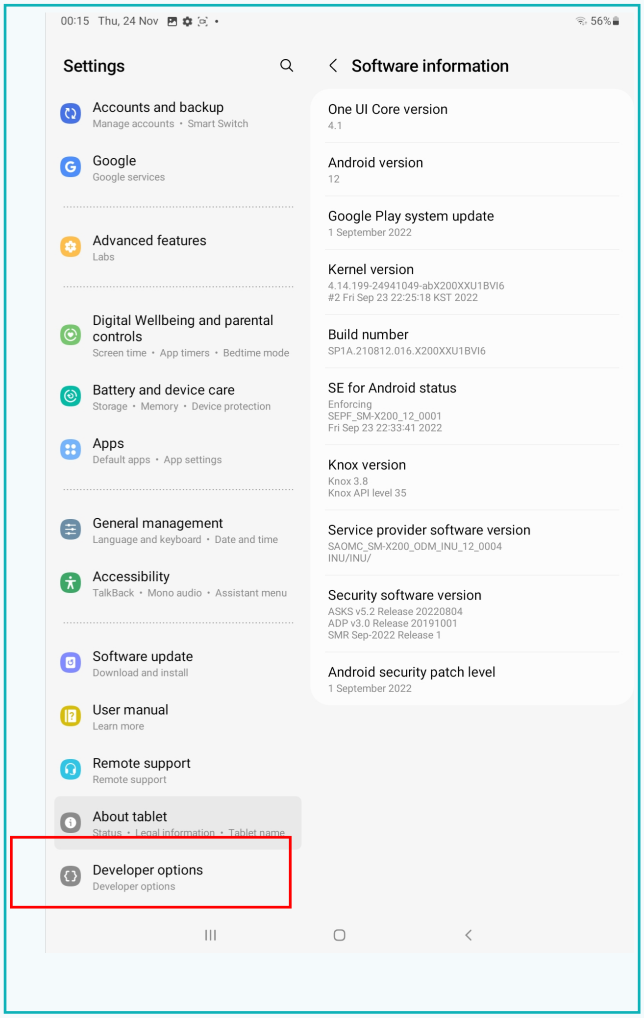
After you navigate to the build number, tap on the build number seven times. A pop-up notification appears, indicating “You are now X steps away from being a developer” appears, with ‘X’ being a number that counts down with every additional tap. On the seventh tap, the ‘Developer’ options are unlocked.

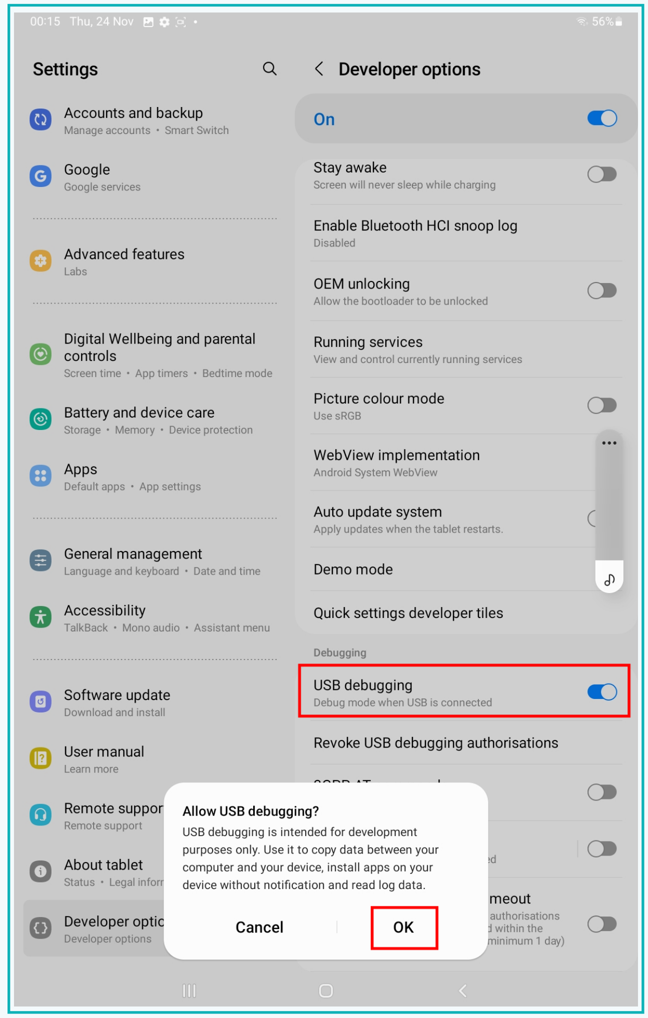




The ‘Developer’ options are now enabled and visible at the bottom of the ‘Settings’ menu.

2 Go to 'Settings' → 'Developer' options (or, if this does not work, on some devices the path is 'Settings' → 'System'→ 'Developer' options) and check the 'USB debugging' checkbox.



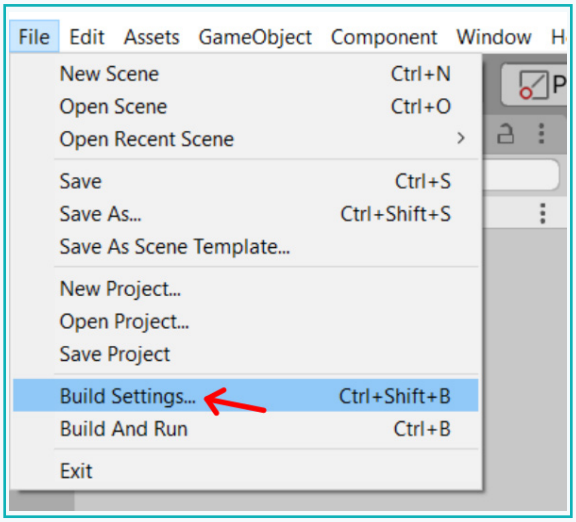


Note:
For different Android versions, the above steps might vary. Follow the instructions for the appropriate Android version.

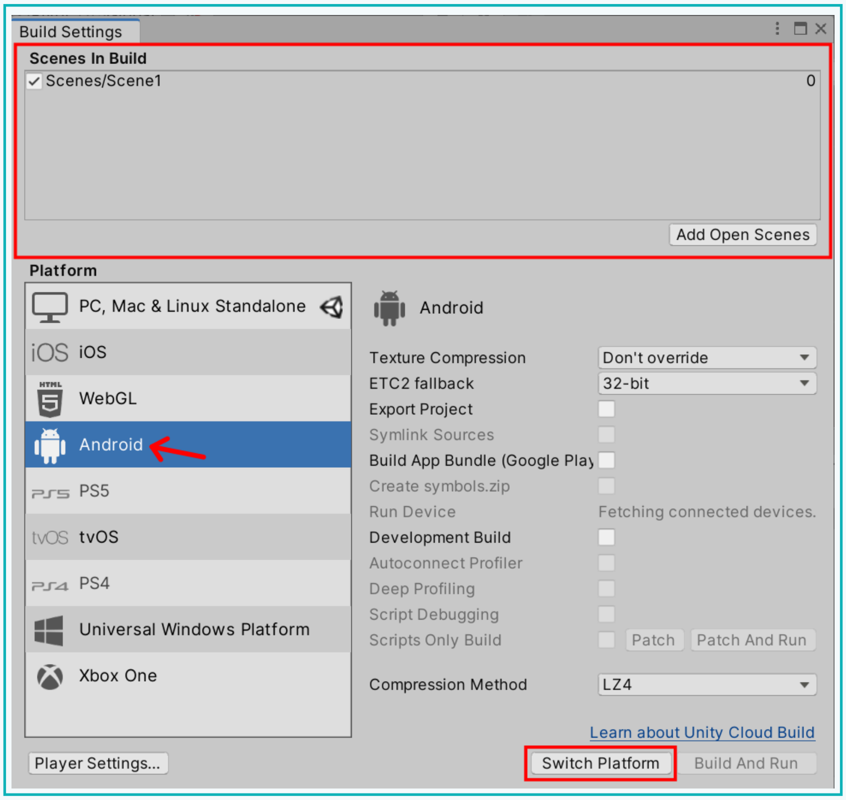
- Next, allow USB debugging on receiving the prompt on the Android device.

We will now change some of the 'Build Settings' within Unity that are required to be able to publish to the Android device.

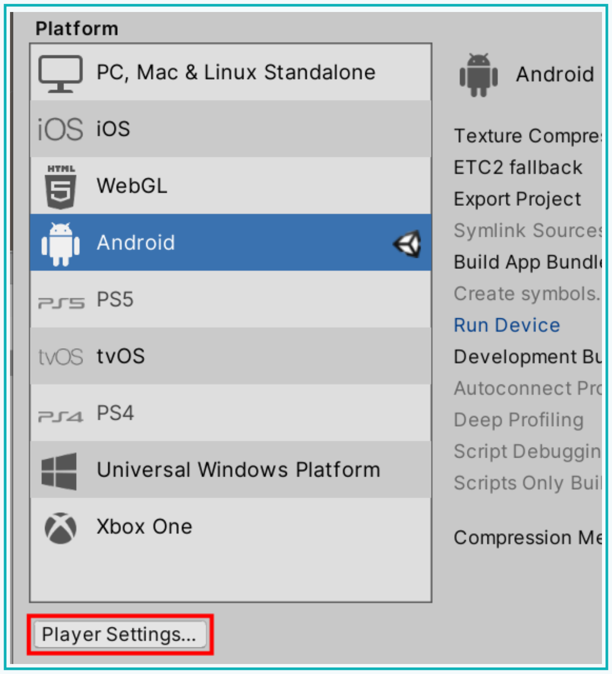
3 Go to ‘File’ → ‘Build Settings’.



- Add the current scene to the ‘Scenes in Build’ list. If there is a scene already present in the list other than what we are working on, right-click and select ‘Remove Selection’. Then, click on ‘Add Open Scenes’ and the active scene will get added to the list.
- Also, select ‘Android’ as the ‘Platform’, and click on the ‘Switch Platform’ button.

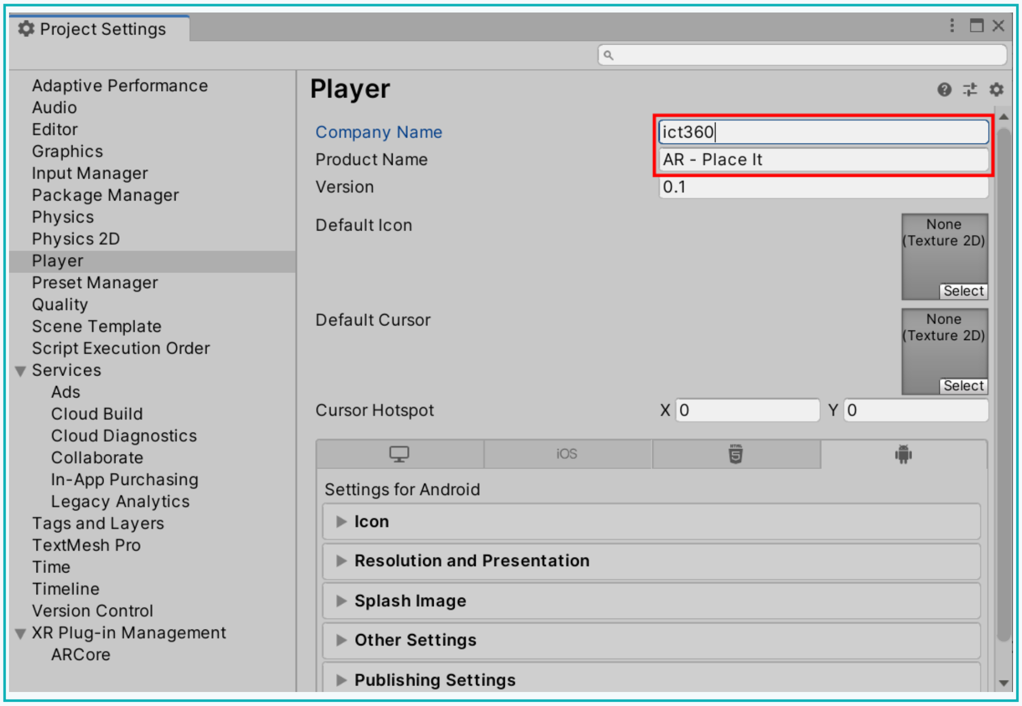


- Next, click on the ‘Player Settings’ button at the bottom left corner of the ‘Build Settings’ window.



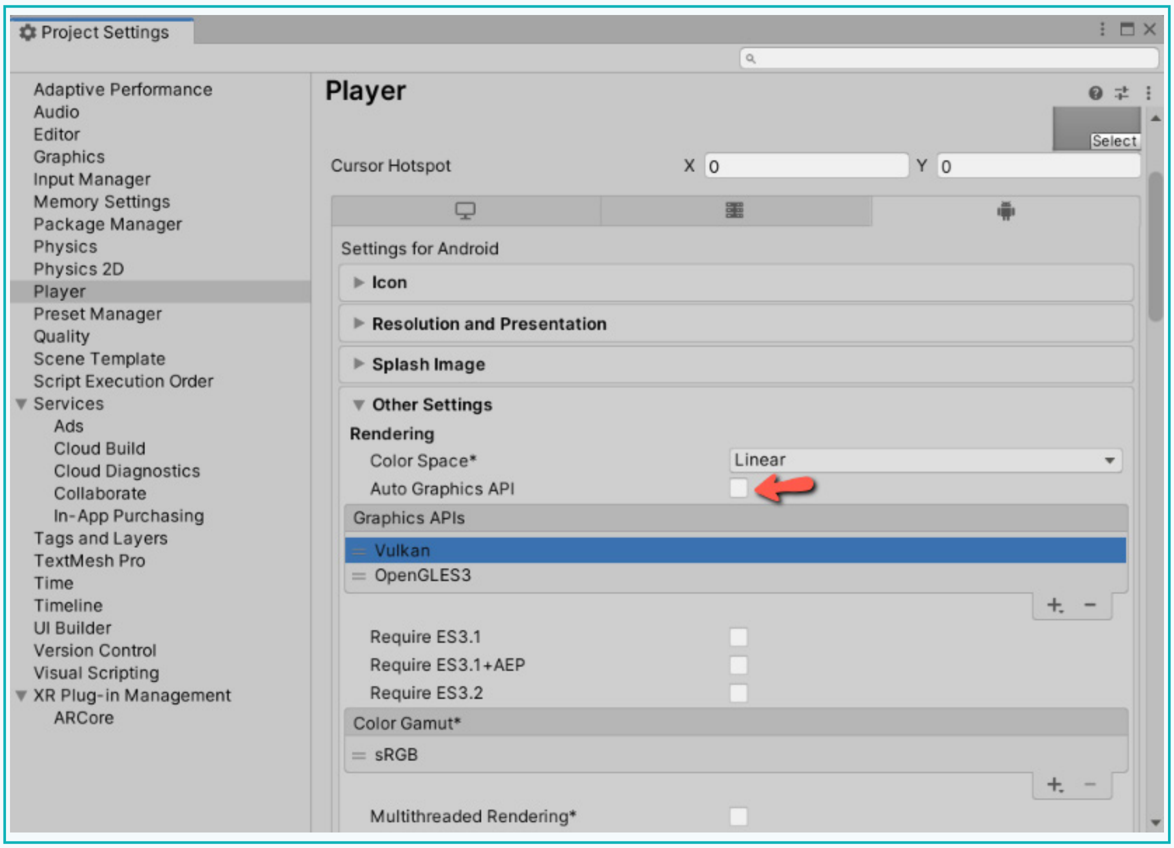
We need to make a few changes here.

a. Company Name and Product Name: Change the company name and product name.



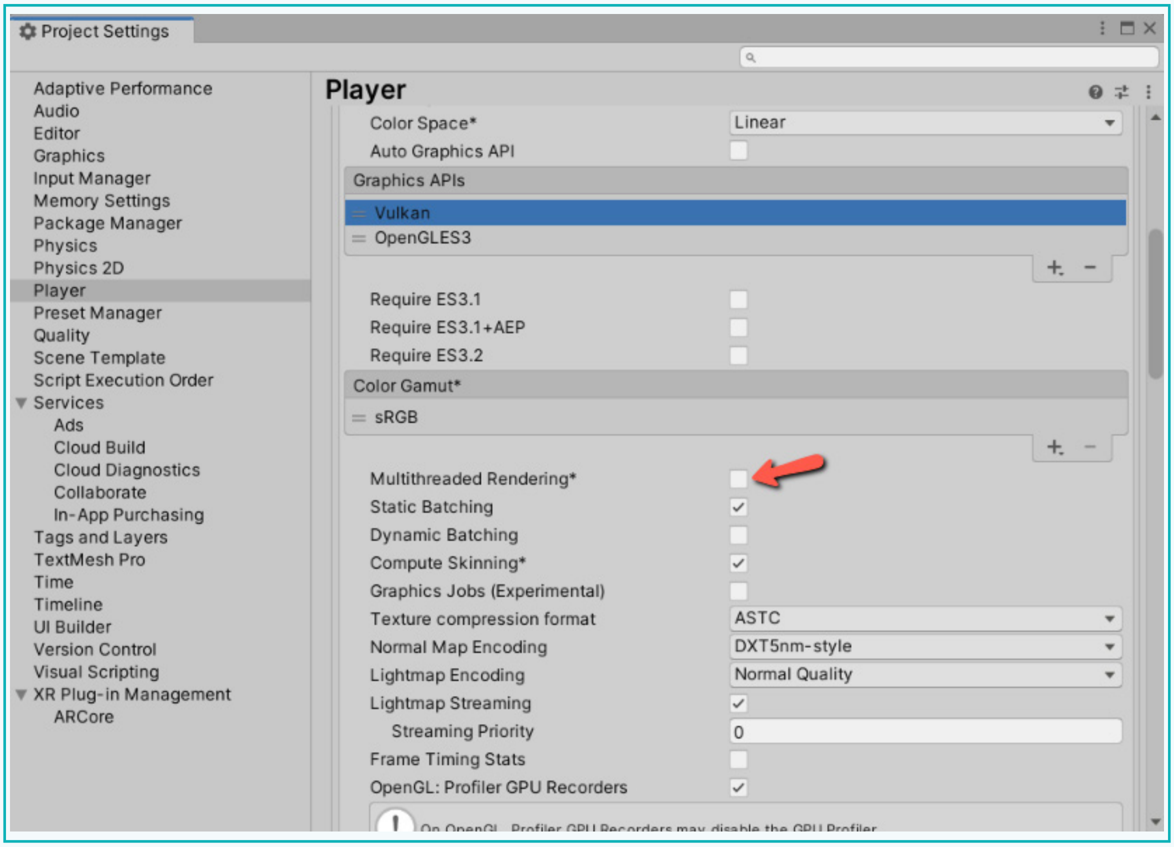
b. Uncheck ‘Auto Graphics APIs’. From the list that appears below, select ‘Vulkan’ and then click on the ‘-‘ button below it.

Vulkan is faster than OpenGL, but not all devices support Vulkan. To be accessible on maximum devices, we are removing Vulkan from the ‘Graphics’ API list.



c. Disable 'Multithreaded rendering'. AR on Android works best when multithreaded rendering is disabled.

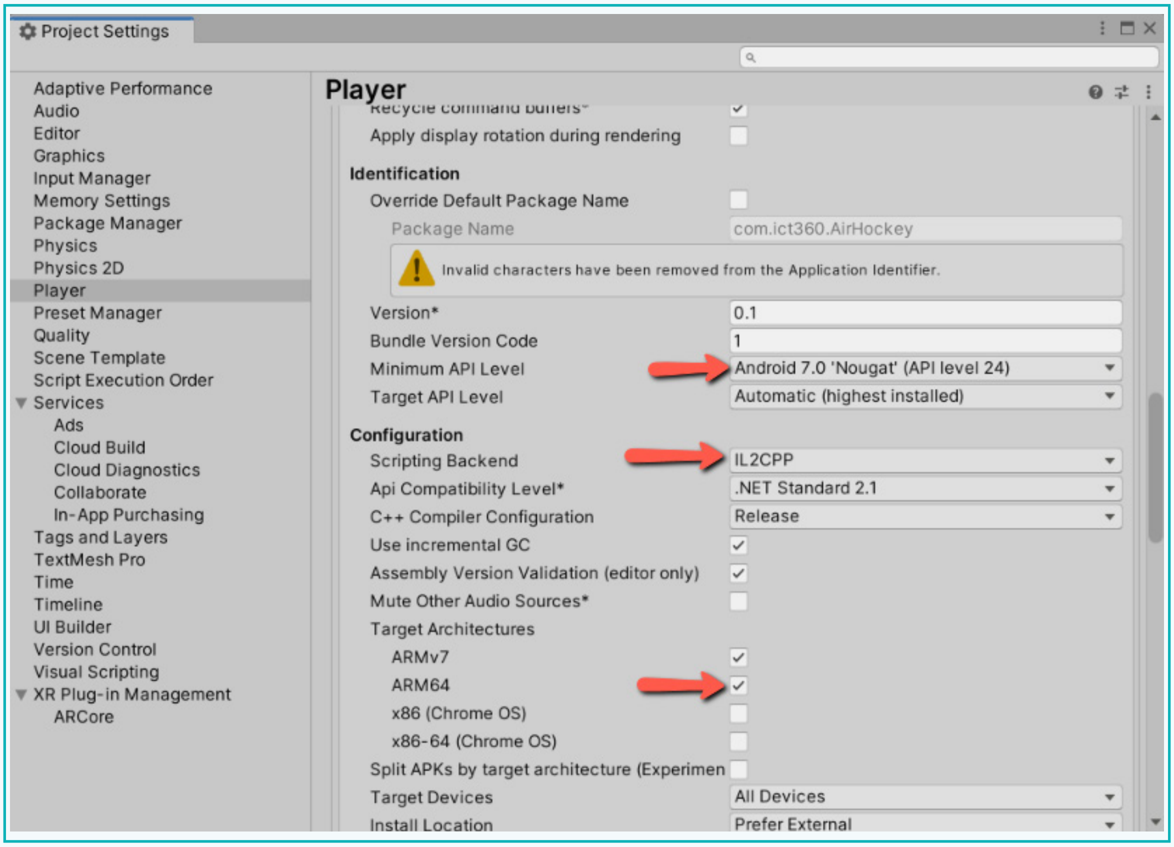
Multithreaded rendering means that the CPU will use multiple threads for rendering. A thread allows multiple processes to execute at the same time, improving efficiency. It helps in improving performance in areas where CPU usage is high. However, multithreaded rendering is unstable for some Android devices. Hence, we keep it off.



d. Minimum API Level – Set to the minimum API level ‘24’, which is required for ARCore to work.

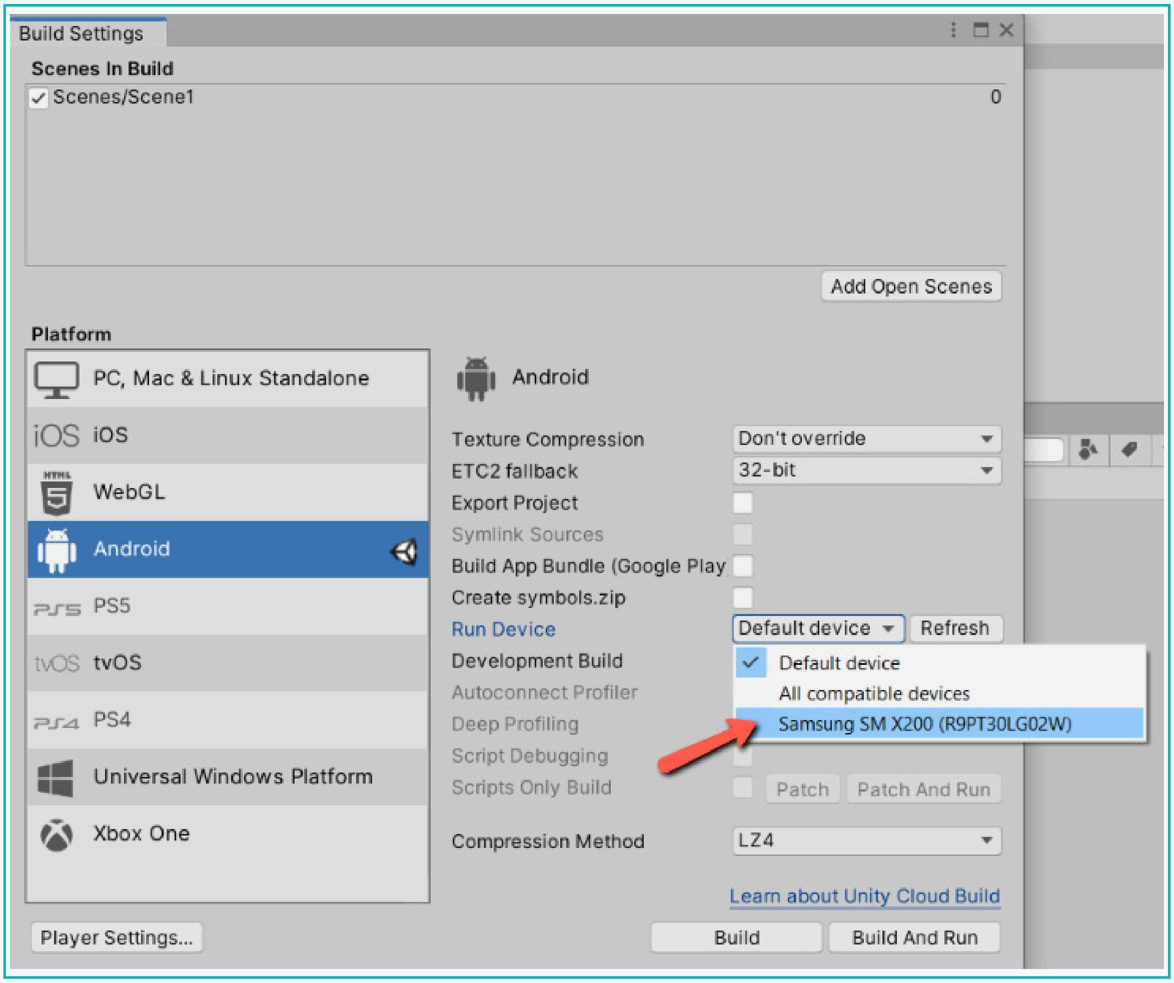
e. Scripting Backend – IL2CPP – gives a performance boost.

f. Target Architectures – Set to ‘ARM64’ to improve performance boost on ARM64 devices. ARM64 devices can run ARMv7 as well.



4 Close the 'Player' settings and go back to 'Build Settings'. Click on the dropdown next to 'Run Device'. Select the 'Android' device that is connected through a USB cable to the laptop/computer.

If your device does not appear on the list here, it indicates that it is not connected, or 'Developer Options/ USB Debugging' has not been enabled.



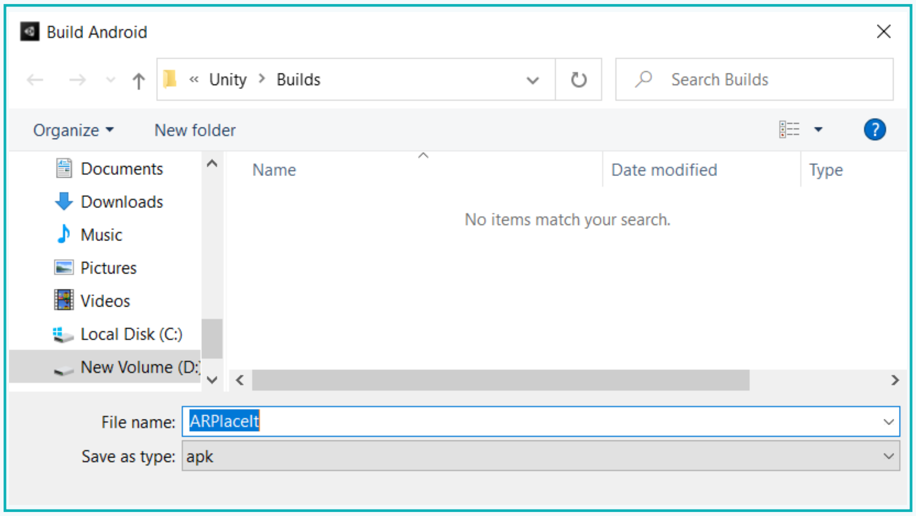
Finally, we will Build and Run our application on the target device.

Note:
There are two ways to create your Build and test it. As observed, there are two buttons – 'Build' & 'Build and Run'.

Build– This option will create a '.apk' file, which is your final build or executable file, and will save it at a specified location on your laptop/computer. Later, you need to connect to an Android device via a USB cable and transfer and install the '.apk' file on the Android device.

Build And Run– This option will directly create the '.apk' file and run it on the device that is connected to the laptop/computer.

5. Test by clicking on 'File' → 'Build and Run'. It will open the 'Save' dialog box and prompt you to save the '.apk' file at a location in your directory. Name it and click on 'Save'. It will build your application and run it directly on the connected Android device.



Once Unity launches on the Android device, allow permission for your AR Application to use your device camera.

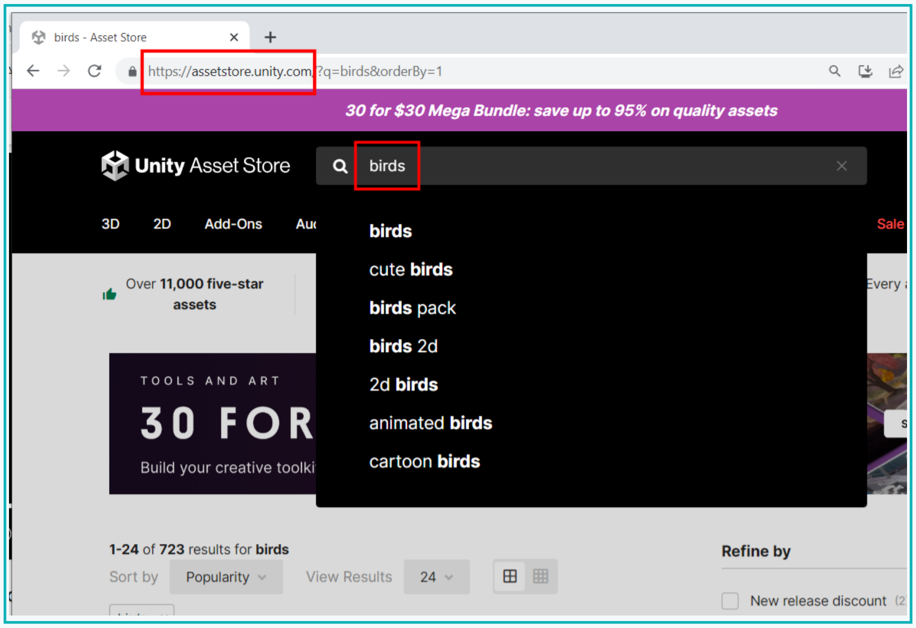
Expected Output– The cube should be visible in your surroundings.

We can make this scene more interesting by adding another object instead of a cube. Let us import an animated 3D model of a bird from the Unity Asset Store and use it in our AR Application.

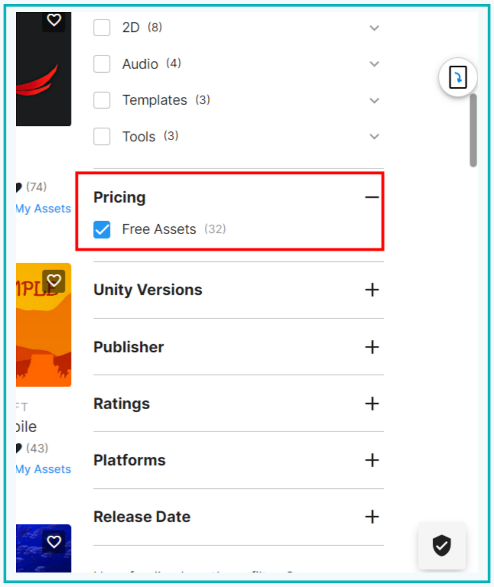
Import Package from Asset Store

Follow the steps given below to import a package of flying birds from the Asset Store.

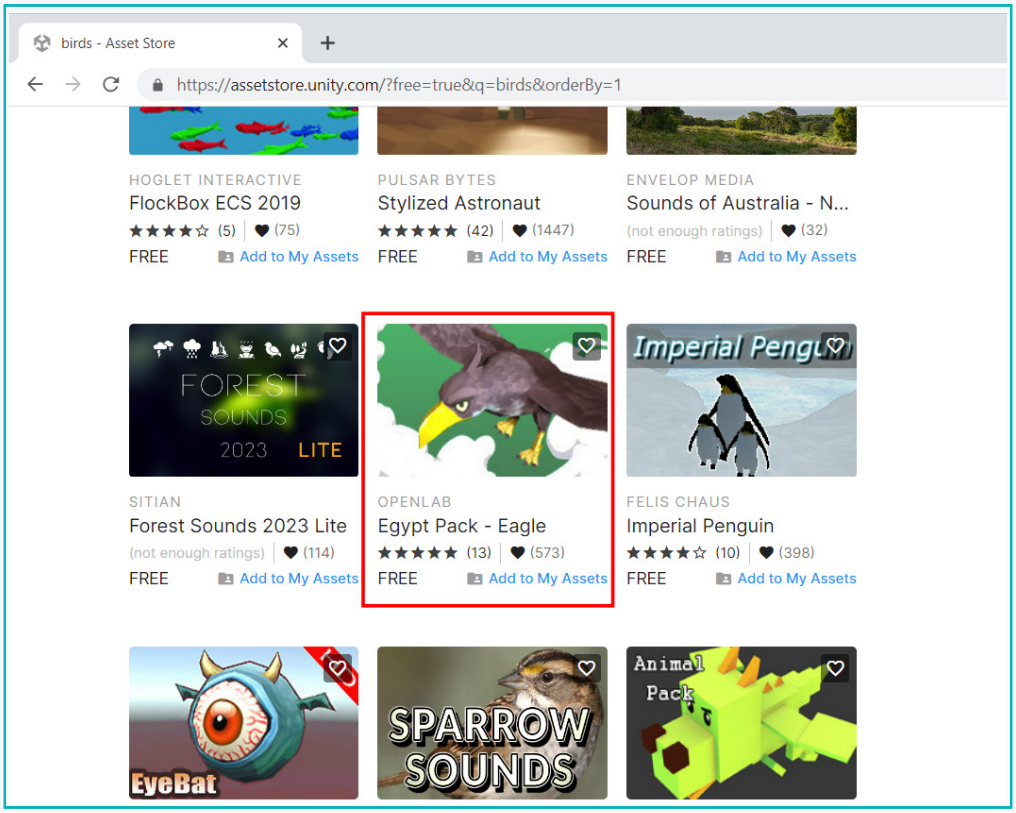
1. Go to https://assetstore.unity.com. In the Search bar, type "birds".



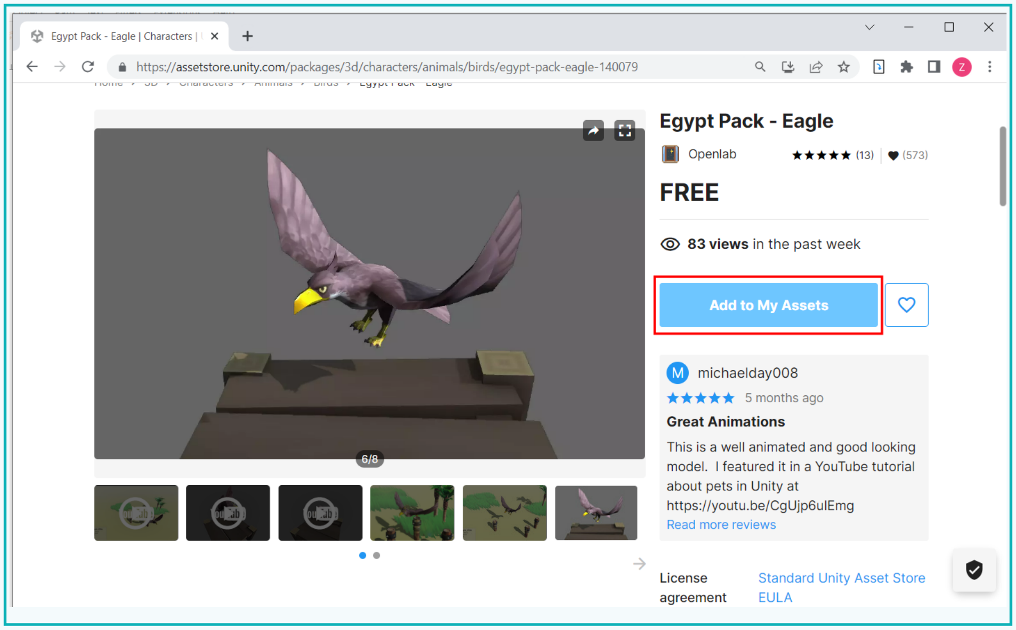
2. Under 'Pricing', check the 'Free Assets' option.



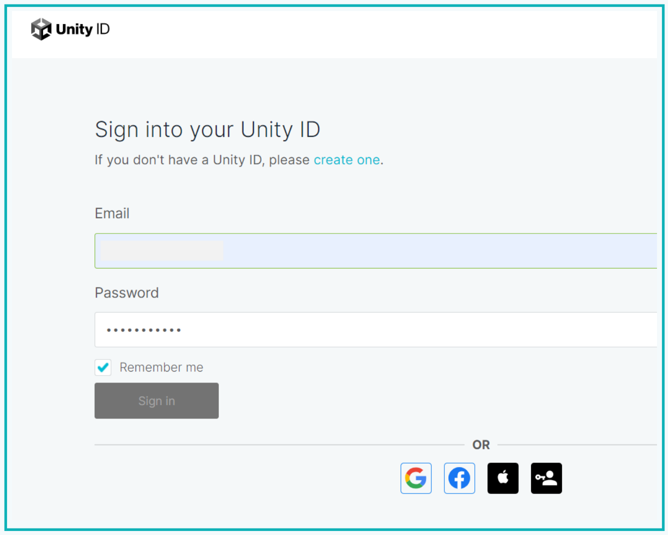
3. Select the ‘Egypt Pack – Eagle’ from the ‘Search’ list.



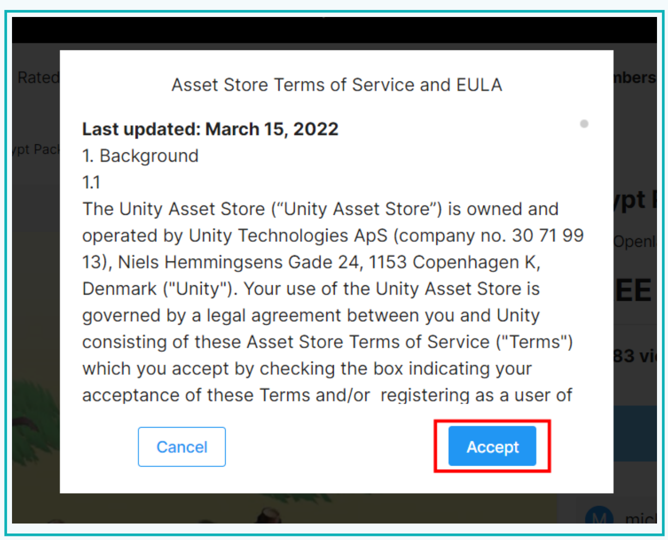
4. Click on the ‘Add to My Assets’ button.



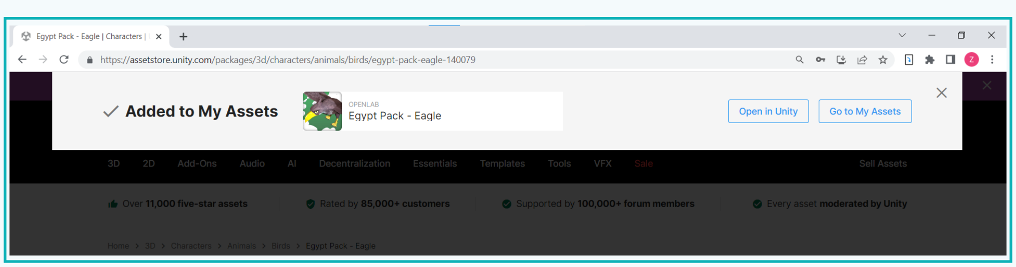
5 At this stage, you may be prompted to log in to Unity if you have not already done so. Please use the same credentials as you did during the Unity installation process.



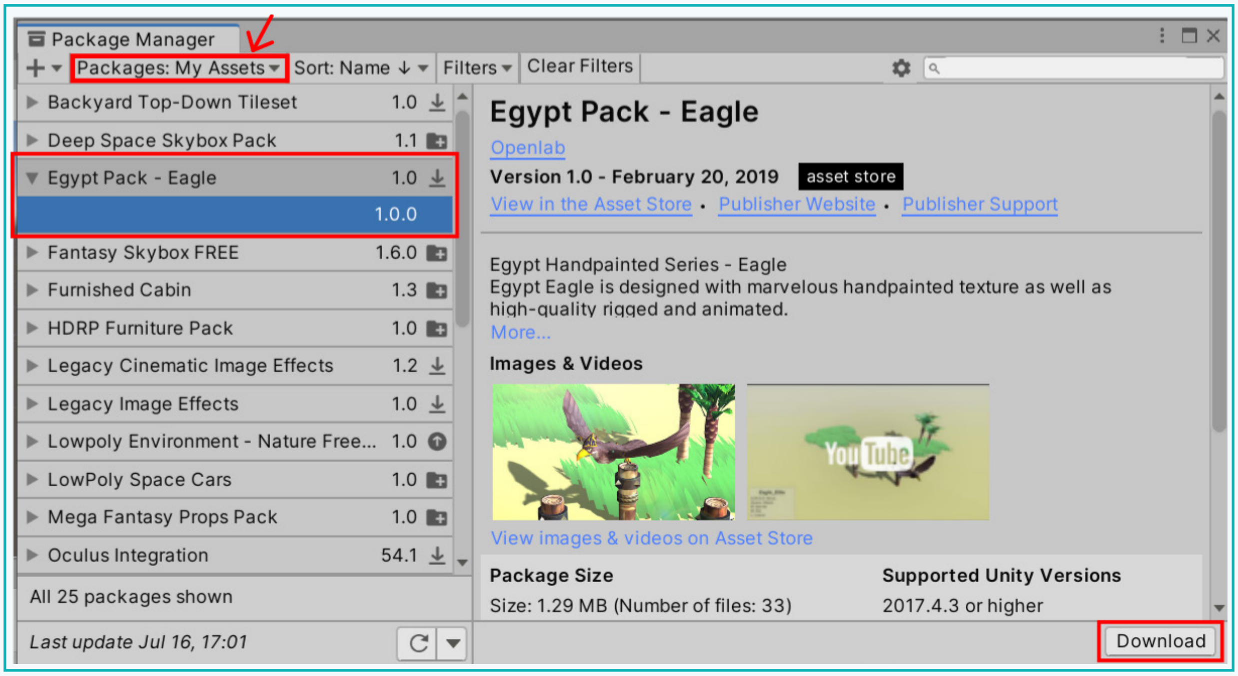
a. Click on 'Accept'.



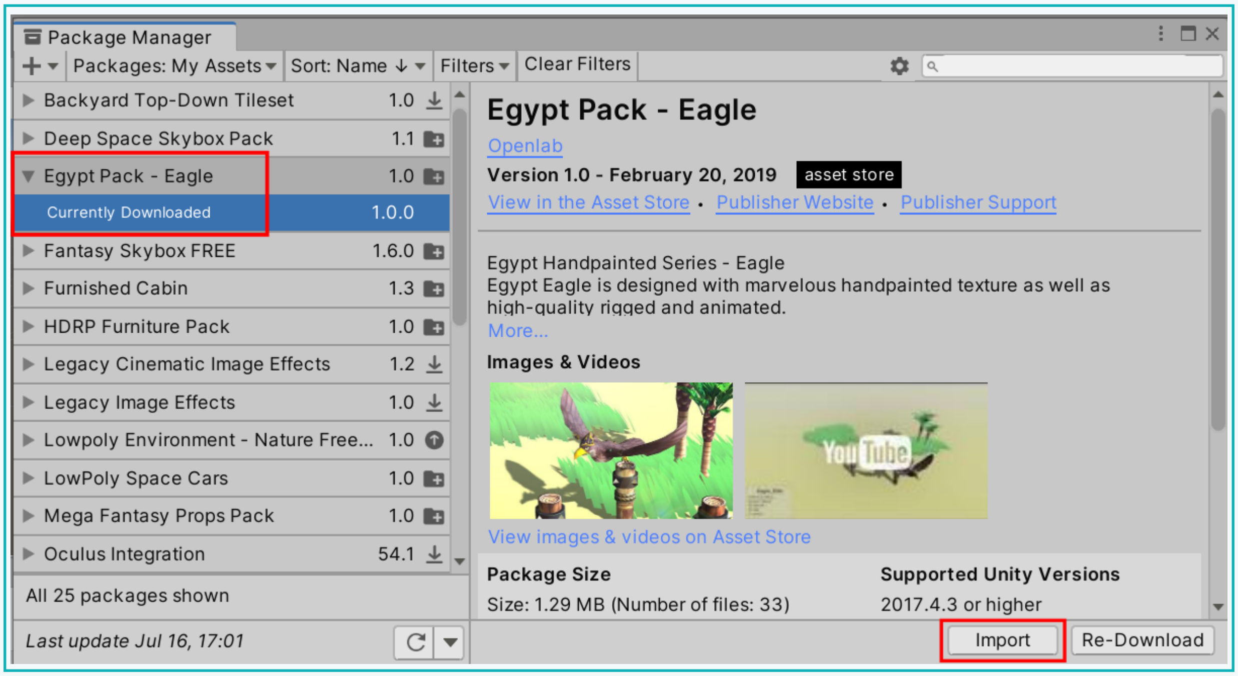
b. A message will be displayed at the top of the page – "Added to My Assets".



6 Now, go back into the Package Manager within the Unity Editor. Select 'My Assets'. The newly imported package will be added to the list. Select it and click on the 'Download' button.

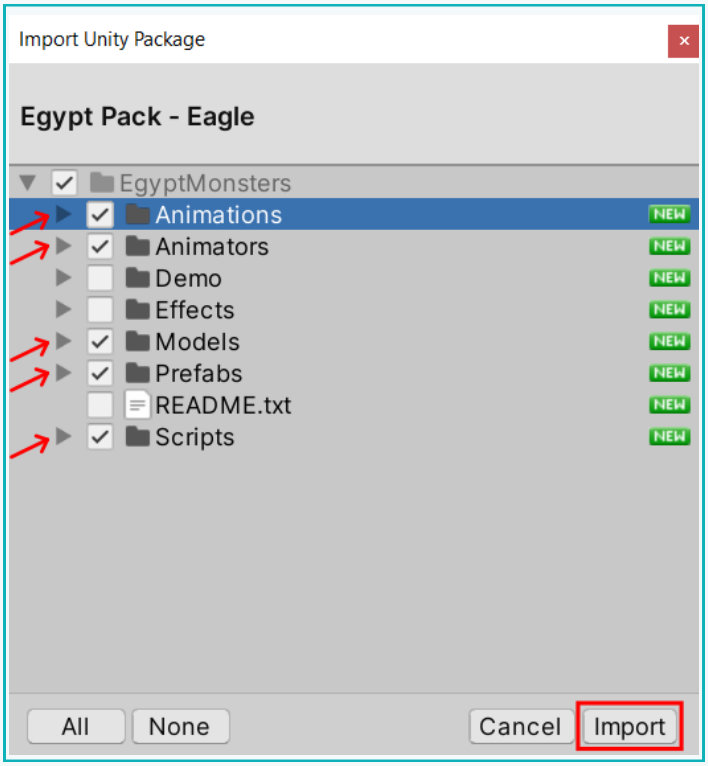


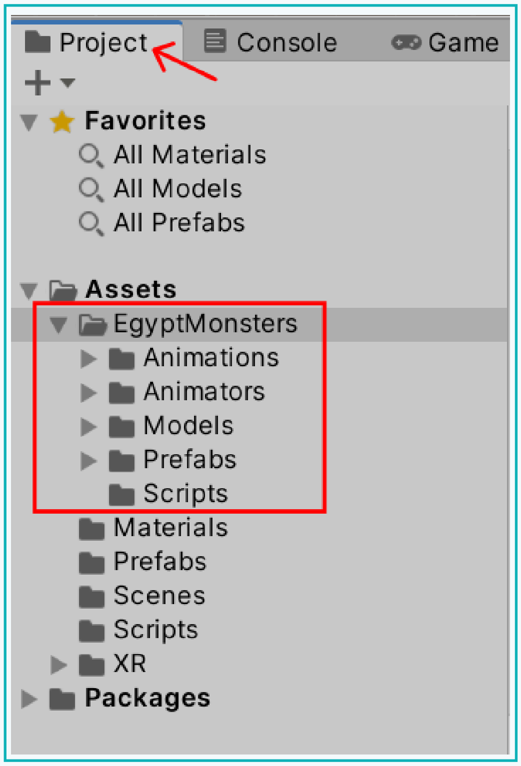
7 Once downloaded, click on the 'Import' button to import the package into your project.



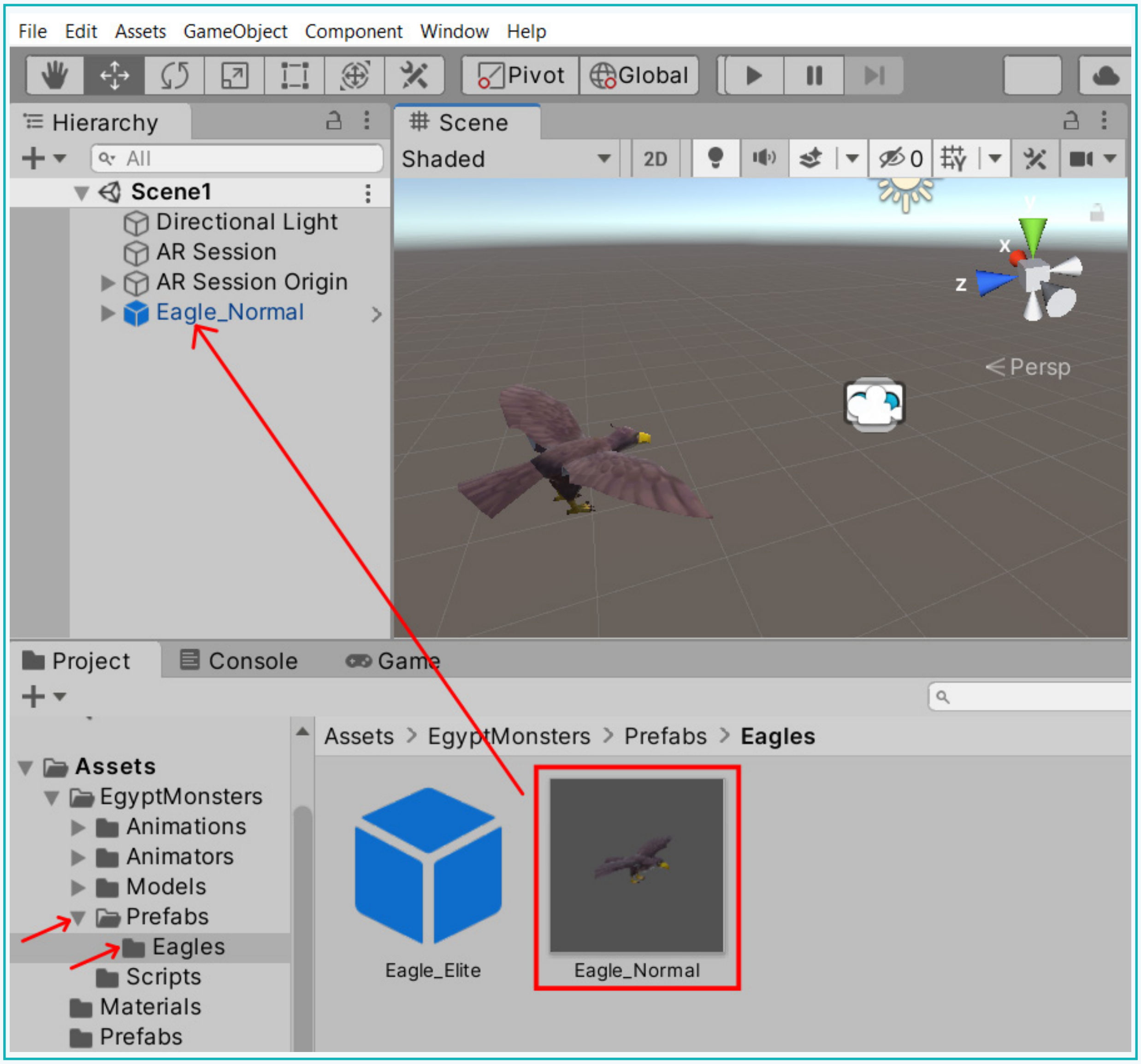
8 In the ‘Import’ dialog box, you can choose which assets from the package you want to import into your project. Let us select the assets that are highlighted in the image. Click on the ‘Import’ button.

The package containing the selected components is imported into the ‘Assets’ folder.

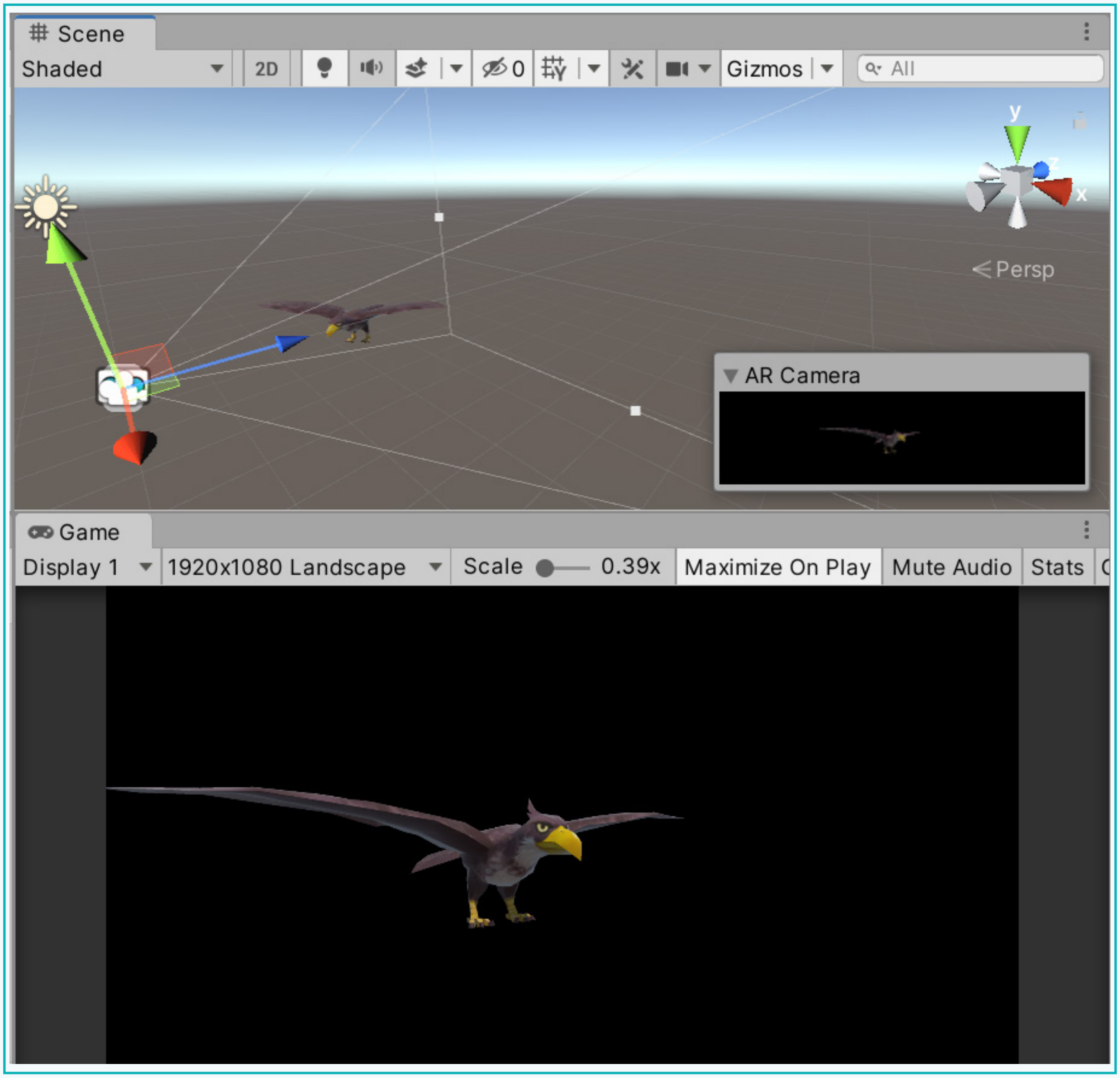




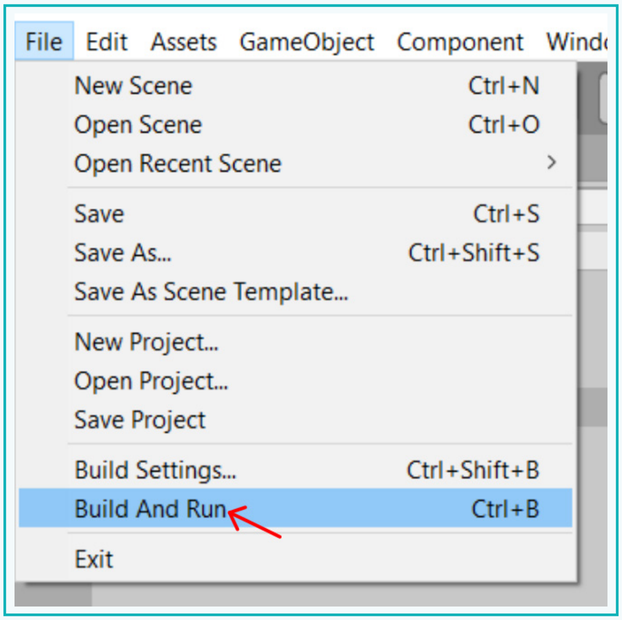
9 Remove the existing 'cube' gameobject from the scene. Navigate to the 'Prefabs' → 'Eagles' folder. Drag and drop 'Eagles_Normal.fbx' into the hierarchy window.



10 Position the prefab within the camera view, as desired.



11 Finally, click on File → 'Build And Run'.



Expected Output

Exercise

Fill in the blanks.

1. To publish XR-based applications to the Android platform, we need to install the __________________ support in Unity.

2. Packages can be imported into Unity using the __________________.

3. Using the __________________ package, we can build platform-independent AR Applications.

4. ARCore is developed by __________________ and ARKit is developed by __________________.

5. The __________________ acts as a bridge between the real and virtual worlds.

6. To enable the 'Developer' options on the Android device, we need to tap on the build number __________________ times.

State if the statements are True or False

7. We use the default Main Camera present in the scene for the AR Applications.

▲ ______________________

8. While developing AR applications, installation of packages is optional.

▲ ______________________

9. We must enable the 'Developer' options on the Android phone to be able to test our AR application on it.

▲ ______________________

Exercise

Tick mark ☑ the correct answer

10. _______ overlays virtual objects in our real-world environment.

a. AR	☐	**b.** VR	☐
c. MR	☐	**d.** XR	☐

11. _______ gives the user the feeling of being present in a completely new environment.

a. AR	☐	**b.** VR	☐
c. MR	☐	**d.** XR	☐

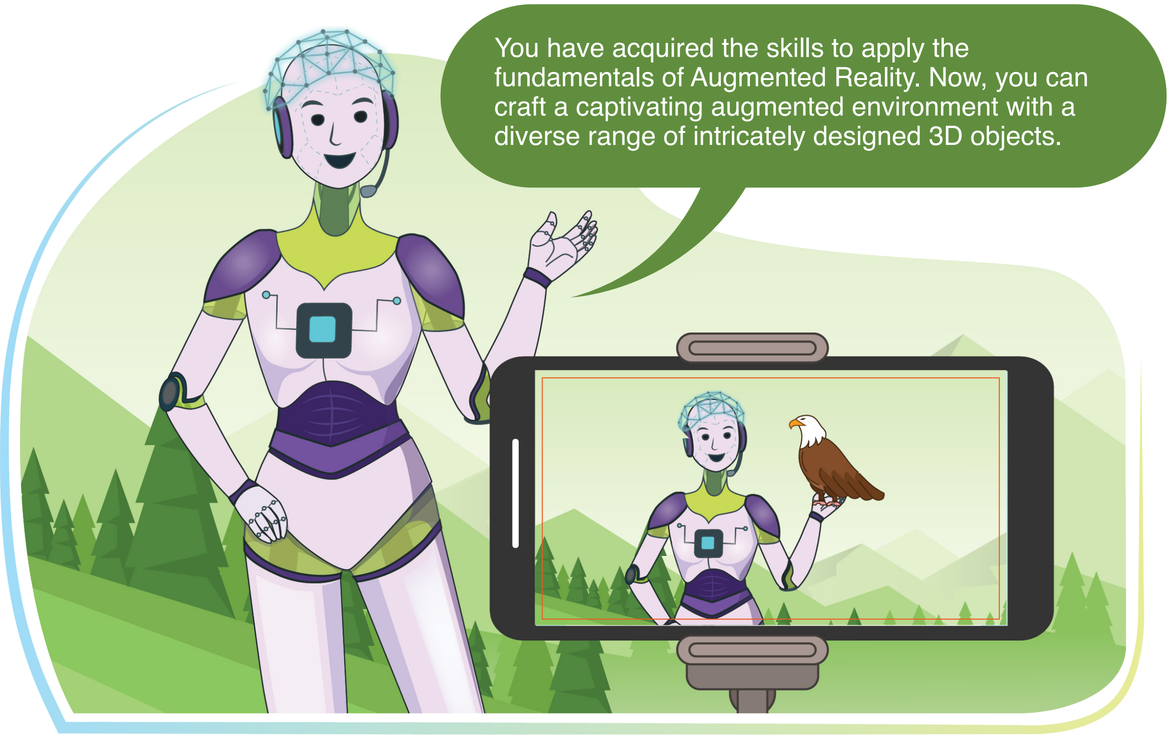
12. _________ is a mixture of both AR & VR.

a. AR	☐	**b.** VR	☐
c. MR	☐	**d.** XR	☐

13. _________ is an umbrella term for all the available immersive technologies.

a. AR	☐	**b.** VR	☐
c. MR	☐	**d.** XR	☐

In this lesson, we understood the need for XR and learnt about the differences between the different immersive technologies. We also learnt about the packages that need to be installed for AR to work and understood their importance in developing a stable AR application. Finally, we developed and published our first AR Application to an Android device.



Tech Flash

- **Extended Reality or XR,** is an umbrella term for all the available immersive technologies like AR (Augmented Reality), VR (Virtual Reality), and MR (Mixed Reality).
- **Augmented Reality** overlays virtual objects in our real-world environment.
- **Virtual Reality** provides the user with the feeling of being present in a completely new environment. VR makes use of computer technology to create a simulated environment that can be explored in 360 degrees.
- **Mixed Reality,** as the name suggests, is a mixture of both AR & VR. In MR, a user can interact with both the real and virtual worlds.
- The **AR Foundation** package is required in Unity AR projects because it enables developers to create AR experiences that are compatible with multiple AR platforms.
- **ARCore** is developed by Google, which provides AR platform-specific functionalities and capabilities for Android devices.
- **ARKit** is developed by Apple, which provides AR platform-specific functionalities and capabilities for iOS devices.
- **AR Session** controls the lifecycle of an AR experience, enabling or disabling AR on the target platform.
- **AR Session Origin** is a crucial component that acts as a bridge between the real and virtual worlds when developing Augmented Reality (AR) applications.